

LVDSL — Pipeline Experimental Integrado

1. Estructura de Salida y Artefactos (Árbol)

Reducción Dimensional (lvdsl_classify_dim_reduction)

```
lvdsl_classify_dim_reduction/  
├─ lvdsl_reduc_dim.txt  
├─ lvdsl_reduc_dim_2026_08_14.img  
│   ├─ plot_Isomap_2D.png  
│   ├─ plot_Isomap_3D.png  
│   ├─ plot_LDA_2D.png  
│   ├─ plot_LDA_3D.png  
│   ├─ plot_LLE_standard_2D.png  
│   ├─ plot_LLE_standard_3D.png  
│   ├─ plot_NCA_2D.png  
│   ├─ plot_NCA_3D.png  
│   ├─ plot_PCA_2D.png  
│   ├─ plot_PCA_3D.png  
│   ├─ plot_SparsePCA_2D.png  
│   ├─ plot_SparsePCA_3D.png  
│   ├─ plot_TruncatedSVD_2D.png  
│   └─ plot_TruncatedSVD_3D.png  
├─ lvdsl_reduc_dim_2026_08_15.ds  
│   ├─ dataset_Isomap_2D.csv  
│   ├─ dataset_Isomap_3D.csv  
│   ├─ dataset_LDA_2D.csv  
│   ├─ dataset_LDA_3D.csv  
│   ├─ dataset_LLE_standard_2D.csv  
│   ├─ dataset_LLE_standard_3D.csv  
│   ├─ dataset_NCA_2D.csv  
│   ├─ dataset_NCA_3D.csv  
│   ├─ dataset_PCA_2D.csv  
│   ├─ dataset_PCA_3D.csv  
│   ├─ dataset_SparsePCA_2D.csv  
│   ├─ dataset_SparsePCA_3D.csv  
│   └─ dataset_TruncatedSVD_2D.csv
```

Rejilla de Clasificación (lvdsl_classify_grid)

```
lvdsl_classify_grid/
├─ lvdsl_classify_20260815.model
│   ├─ model_DecisionTree_LDA_2D.model
│   ├─ model_DecisionTree_LDA_3D.model
│   ├─ model_DecisionTree_NCA_2D.model
│   ├─ model_DecisionTree_NCA_3D.model
│   ├─ model_DecisionTree_Original_Scaled.model
│   ├─ model_DecisionTree_PCA_2D.model
│   ├─ model_DecisionTree_PCA_3D.model
│   ├─ model_DecisionTree_SparsePCA_2D.model
│   ├─ model_DecisionTree_TruncatedSVD_2D.model
│   ├─ model_DecisionTree_TruncatedSVD_3D.model
│   ├─ model_KNN_LDA_2D.model
│   ├─ model_KNN_LDA_3D.model
│   ├─ model_KNN_NCA_2D.model
│   ├─ model_KNN_NCA_3D.model
│   ├─ model_KNN_Original_Scaled.model
│   ├─ model_KNN_PCA_2D.model
│   ├─ model_KNN_PCA_3D.model
│   ├─ model_KNN_TruncatedSVD_2D.model
│   ├─ model_KNN_TruncatedSVD_3D.model
│   ├─ model_LogisticRegression_LDA_2D.model
│   ├─ model_LogisticRegression_LDA_3D.model
│   ├─ model_LogisticRegression_NCA_2D.model
│   ├─ model_LogisticRegression_NCA_3D.model
│   ├─ model_LogisticRegression_Original_Scaled.model
│   ├─ model_LogisticRegression_PCA_2D.model
│   ├─ model_LogisticRegression_PCA_3D.model
│   ├─ model_LogisticRegression_TruncatedSVD_2D.model
│   ├─ model_LogisticRegression_TruncatedSVD_3D.model
│   ├─ model_RandomForest_LDA_2D.model
│   └─ model_RandomForest_LDA_3D.model
├─ model_RandomForest_NCA_2D.model
├─ model_RandomForest_NCA_3D.model
├─ model_RandomForest_Original_Scaled.model
├─ model_RandomForest_PCA_2D.model
├─ model_RandomForest_PCA_3D.model
├─ model_RandomForest_TruncatedSVD_2D.model
├─ model_RandomForest_TruncatedSVD_3D.model
├─ model_XGBoost_LDA_2D.model
├─ model_XGBoost_LDA_3D.model
├─ model_XGBoost_NCA_2D.model
└─ model_XGBoost_NCA_3D.model
=====
LVDSL - BITACORA DE SELECCION Y DATASETS PROCESADOS
FECHA DE EJECUCION: 2026-08-15 22:51:25
```

2. Bitácoras Técnicas de Ejecución

lvdsl_reduc_dim.txt

```
=====
```

```
>>> DIRECTORIOS DE SALIDA GENERADOS <<<
```

- Imágenes (.png): lvdsl_classify_dim_reduction/lvdsl_reduc_dim_2026_08_15.i
- Datasets (.ds): lvdsl_classify_dim_reduction/lvdsl_reduc_dim_2026_08_15.d

```
>>> RESUMEN DE DATASETS FINALES ALMACENADOS <<<
```

- [LDA_2D] -> dataset_LDA_2D.csv | Dimensiones: (169528, 3)
- [NCA_2D] -> dataset_NCA_2D.csv | Dimensiones: (8000, 3)
- [PCA_2D] -> dataset_PCA_2D.csv | Dimensiones: (169528, 2)
- [SparsePCA_2D] -> dataset_SparsePCA_2D.csv | Dimensiones: (8000, 2)
- [TruncatedSVD_2D] -> dataset_TruncatedSVD_2D.csv | Dimensiones: (169528, 2)
- [Isomap_2D] -> dataset_Isomap_2D.csv | Dimensiones: (8000, 2)
- [LLE_standard_2D] -> dataset_LLE_standard_2D.csv | Dimensiones: (8000, 2)
- [LDA_3D] -> dataset_LDA_3D.csv | Dimensiones: (169528, 4)
- [NCA_3D] -> dataset_NCA_3D.csv | Dimensiones: (8000, 4)
- [PCA_3D] -> dataset_PCA_3D.csv | Dimensiones: (169528, 3)
- [SparsePCA_3D] -> dataset_SparsePCA_3D.csv | Dimensiones: (8000, 3)
- [TruncatedSVD_3D] -> dataset_TruncatedSVD_3D.csv | Dimensiones: (169528, 3)
- [Isomap_3D] -> dataset_Isomap_3D.csv | Dimensiones: (8000, 3)
- [LLE_standard_3D] -> dataset_LLE_standard_3D.csv | Dimensiones: (8000, 3)

```
=====
```

```
>>> APOORTE DE CARACTERÍSTICAS ORIGINALES Y PESOS <<<
```

```
--- MODELO: LDA_2D ---
```

```
Matriz de salida: (169528, 2)
```

```
Varianza explicada acumulada: 0.9890 [0.7877, 0.2013]
```

```
Top 3 Características con mayor peso absoluto por Componente:
```

- Comp 1: AF3 (|w|=1.102), AF5 (|w|=0.440), tau (|w|=0.381)
- Comp 2: AD5 (|w|=1.371), AF3 (|w|=0.927), AH3 (|w|=0.283)

lvdsl_classify_summary.txt

```
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```

```
--- MODELO: NCA_3D ---
```

```
Matriz de salida: (8000, 2)
```

```
Top 3 Características con mayor peso absoluto por Componente:
```

- Comp 1: tau (|w|=1419.969), AF3 (|w|=868.100), s (|w|=372.050)

```
1. INFORMACIÓN DE LOS DATASETS EVALUADOS
```

```
-----
```

```
Dataset: Original_Scaled
```

```
- Filas (Muestras): 169528
```

```
--- MODELO: PCA_2D ---
```

```
Columnas Totales: 8 (Características: 7)
```

```
Matriz de salida: (169528, 2)(target): {0: 42382, 1: 42382, 2: 42382, 3: 42382}
```

```
Varianza explicada acumulada: 0.4767 [0.2888, 0.1879]
```

Dataset: LDA_2D

- Filas (Muestras): 169528
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 42382, 1: 42382, 2: 42382, 3: 42382}

Dataset: NCA_2D

- Filas (Muestras): 8000
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {3: 2000, 1: 2000, 2: 2000, 0: 2000}

Dataset: PCA_2D

- Filas (Muestras): 169528
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 42382, 1: 42382, 2: 42382, 3: 42382}

Dataset: SparsePCA_2D

- Filas (Muestras): 8000
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 8000}

Dataset: TruncatedSVD_2D

- Filas (Muestras): 169528
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 42382, 1: 42382, 2: 42382, 3: 42382}

Dataset: Isomap_2D

- Filas (Muestras): 8000
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 8000}

Dataset: LLE_standard_2D

- Filas (Muestras): 8000
- Columnas Totales: 3 (Características: 2)
- Distribución de Clases (target): {0: 8000}

3. Vista Previa de Datasets Proyectados (.ds)

Dataset: LDA_3D
dataset_isomap_2d.csv head(5)

	comp_1	comp_2
-0.271363		
-2.363378		
-1.531930		
-2.103060		

Dataset: PCA_3D

comp_1	comp_2
-3.648700	-0.155227
0.813621	0.017972

dataset_Isomap_3D.csv `head(5)`

comp_1	comp_2	comp_3
-0.271363	-2.363378	-1.926143
-1.531930	-1.270376	-2.616972
-2.103060	1.940961	0.186005
-3.648700	-0.155227	2.233633
0.813621	0.017972	-0.303813

dataset_LDA_2D.csv `head(5)`

comp_1	comp_2	target
-1.779997	-1.155250	0
-0.443100	-0.351514	0
-0.668227	-0.715281	0
-0.008504	-0.291378	0
-1.373740	-0.082527	0

dataset_LDA_3D.csv head(5)

comp_1	comp_2	comp_3	target
-1.779997	-1.155250	0.063141	0
-0.443100	-0.351514	-0.057553	0
-0.668227	-0.715281	-0.099954	0
-0.008504	-0.291378	-0.301185	0
-1.373740	-0.082527	1.525006	0

dataset_LLE_standard_2D.csv head(5)

comp_1	comp_2
0.011244	0.015688
0.011244	0.015637
0.011244	-0.012994
0.011244	-0.015369
0.011244	0.002995

dataset_LLE_standard_3D.csv head(5)

comp_1	comp_2	comp_3
-0.011244	-0.015688	0.006417
-0.011244	-0.015637	-0.022983
-0.011244	0.012994	-0.006726
-0.011244	0.015369	0.009640

comp_1	comp_2	comp_3
-0.011244	-0.002995	-0.003681

dataset_NCA_2D.csv `head(5)`

comp_1	comp_2	target
901.452699	1466.673119	3
-614.946223	2675.300469	3
-1653.510396	-2332.033180	1
-2006.228176	-678.225105	2
-221.779355	-322.062981	0

dataset_NCA_3D.csv `head(5)`

comp_1	comp_2	comp_3	target
229.477273	301.060350	-47.132890	3
57.228648	380.398576	-388.722714	3
-421.244561	-383.857880	159.418285	1
-584.072681	189.567172	467.883073	2
16.168297	-153.016321	-149.504947	0

dataset_PCA_2D.csv head(5)

comp_1	comp_2
0.976867	-0.569523
0.677115	0.400179
-0.152786	-0.983418
-0.095701	0.031133
3.330242	1.049078

dataset_PCA_3D.csv head(5)

comp_1	comp_2	comp_3
0.976867	-0.569523	-1.306374
0.677115	0.400179	-0.415667
-0.152786	-0.983418	-1.238795
-0.095701	0.031133	-0.543433
3.330242	1.049078	1.244720

dataset_SparsePCA_2D.csv head(5)

comp_1	comp_2
-0.420354	0.822620
0.165050	1.243429
1.557836	-0.602323
2.461052	-0.891141

comp_1	comp_2
-0.469006	0.099045

dataset_SparsePCA_3D.csv head(5)

comp_1	comp_2	comp_3
-0.405389	0.984227	-0.388892
0.117555	1.041854	0.777152
1.536224	-0.836558	0.848718
2.592496	-0.115745	-1.941558
-0.489031	-0.014628	0.248811

dataset_TruncatedSVD_2D.csv head(5)

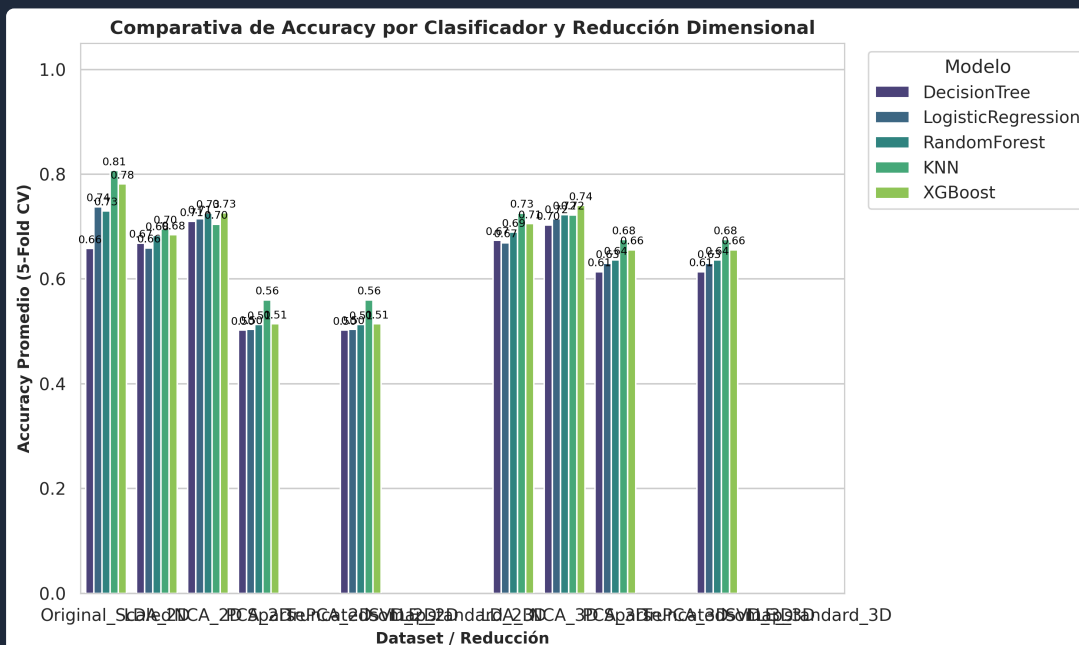
comp_1	comp_2
0.976867	-0.569523
0.677115	0.400179
-0.152786	-0.983418
-0.095701	0.031133
3.330242	1.049078

head(5)

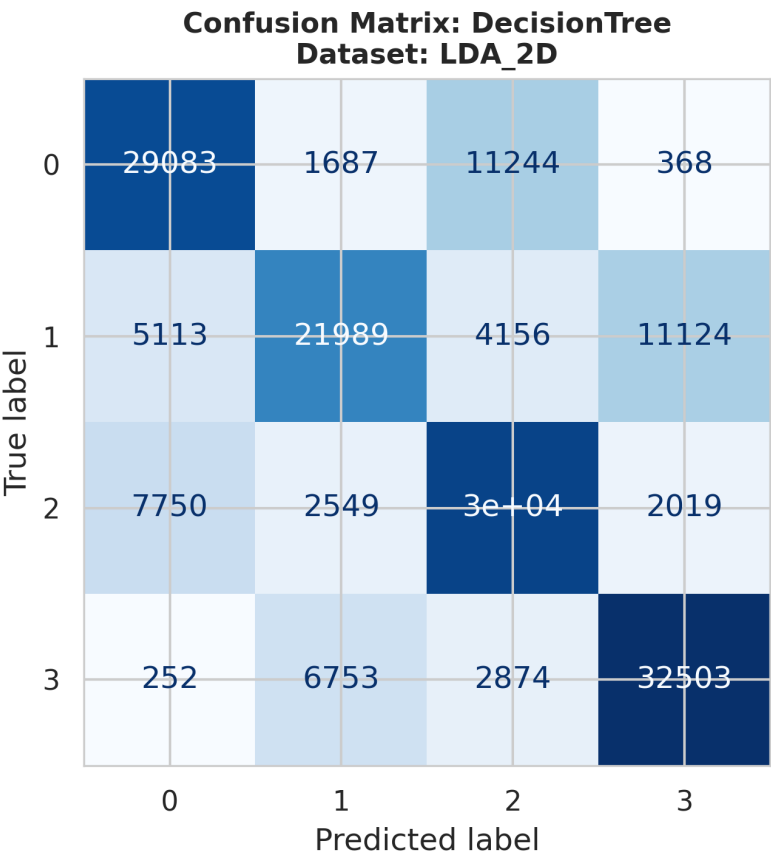
comp_1	comp_2	comp_3
0.976867	-0.569523	-1.306374
0.677115	0.400179	-0.415667
-0.152786	-0.983418	-1.238795
-0.095701	0.031133	-0.543433
3.330242	1.049078	1.244720

4. Resultados de Clasificación (Plots & Matrices)

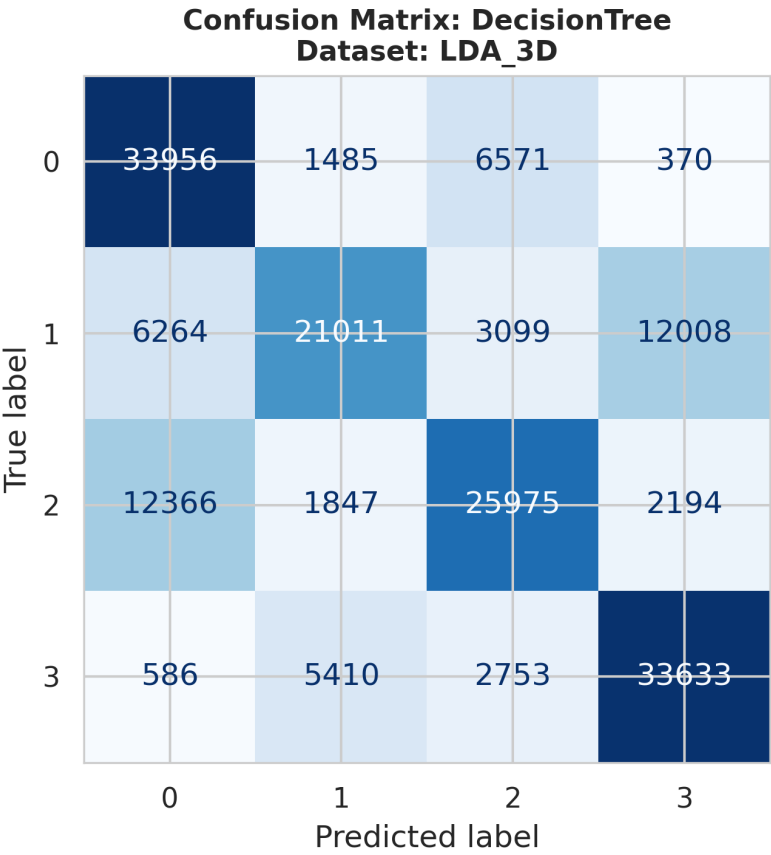
Accuracy_Comparison_All_Models.png



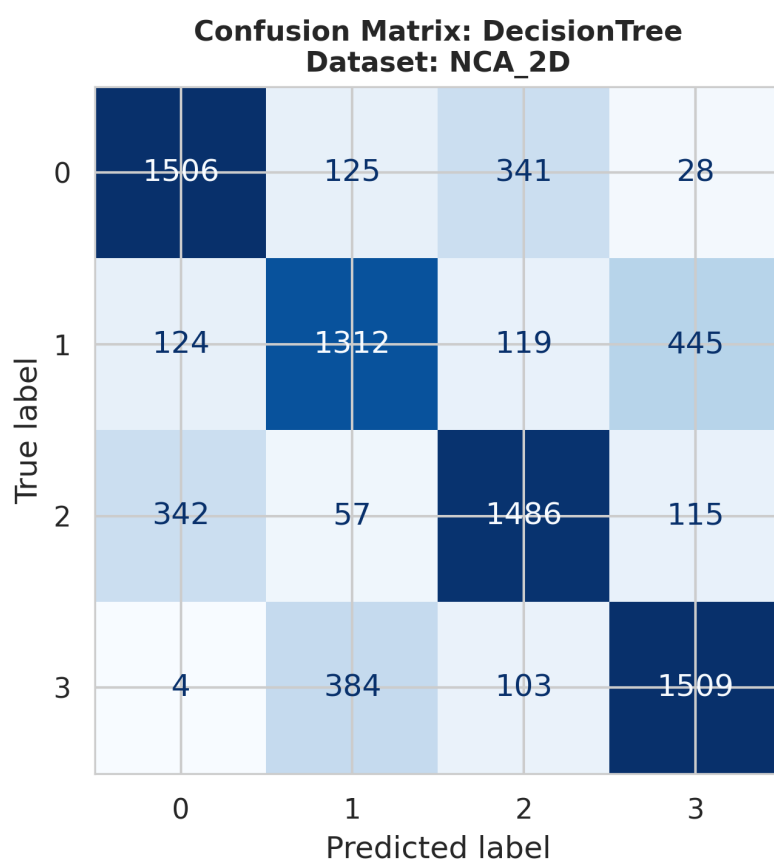
ConfusionMatrix_DecisionTree_LDA_2D.png



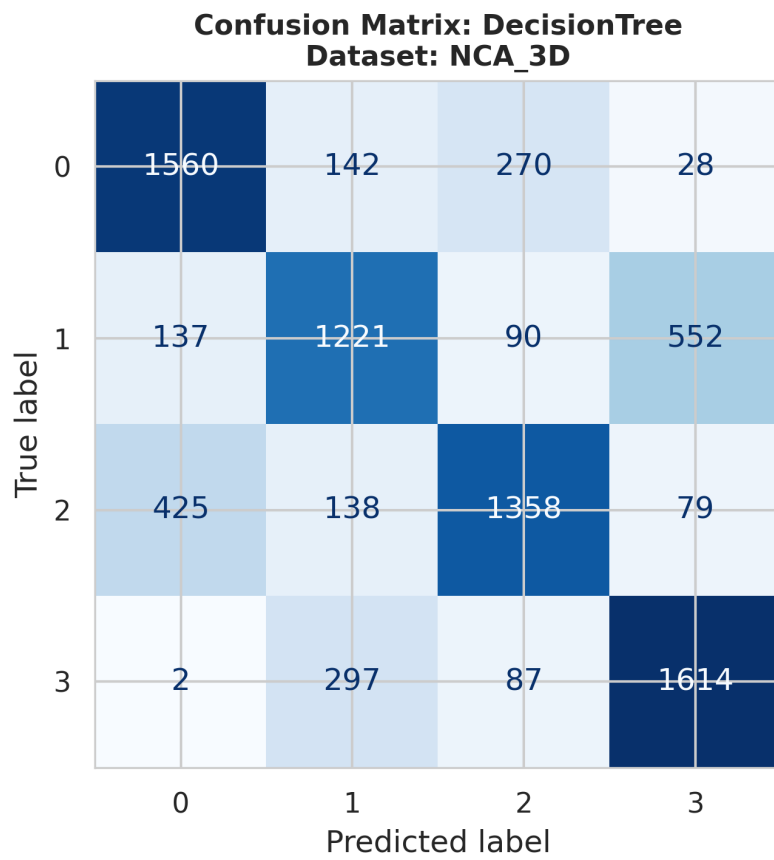
ConfusionMatrix_DecisionTree_LDA_3D.png



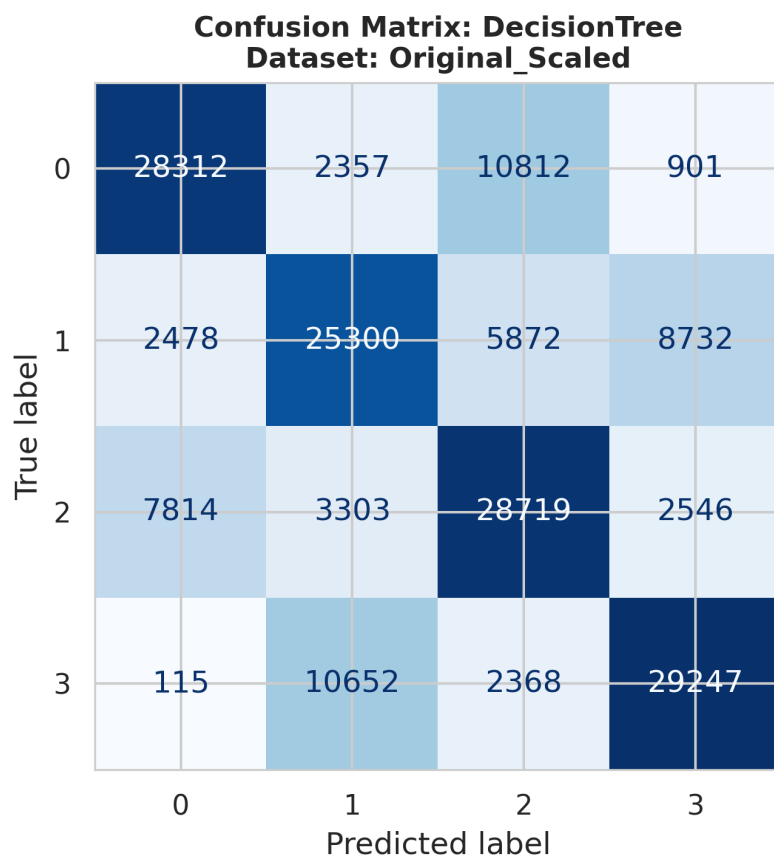
ConfusionMatrix_DecisionTree_NCA_2D.png



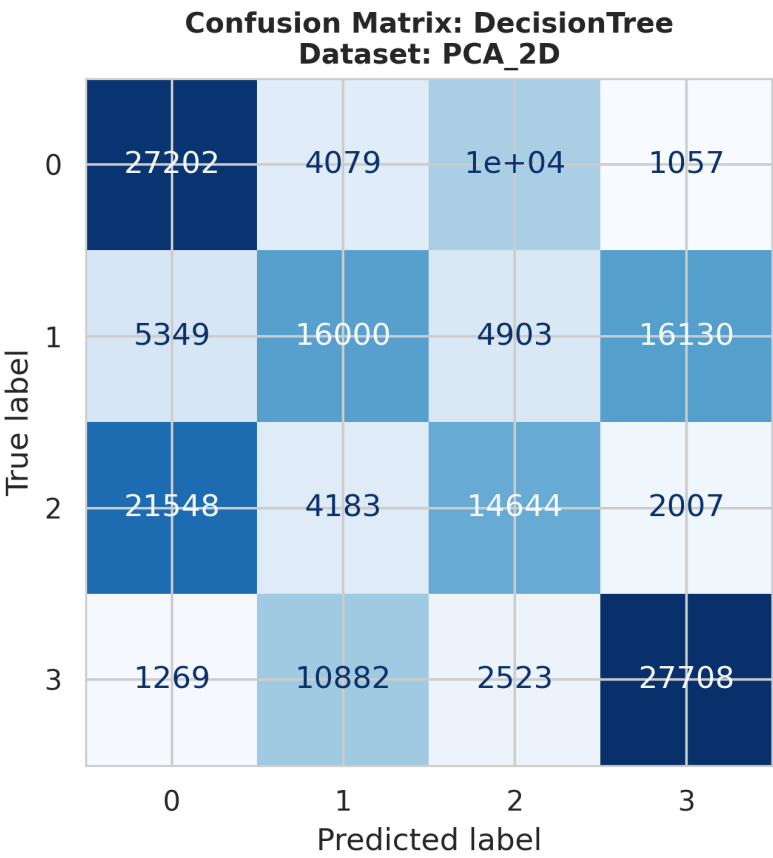
ConfusionMatrix_DecisionTree_NCA_3D.png



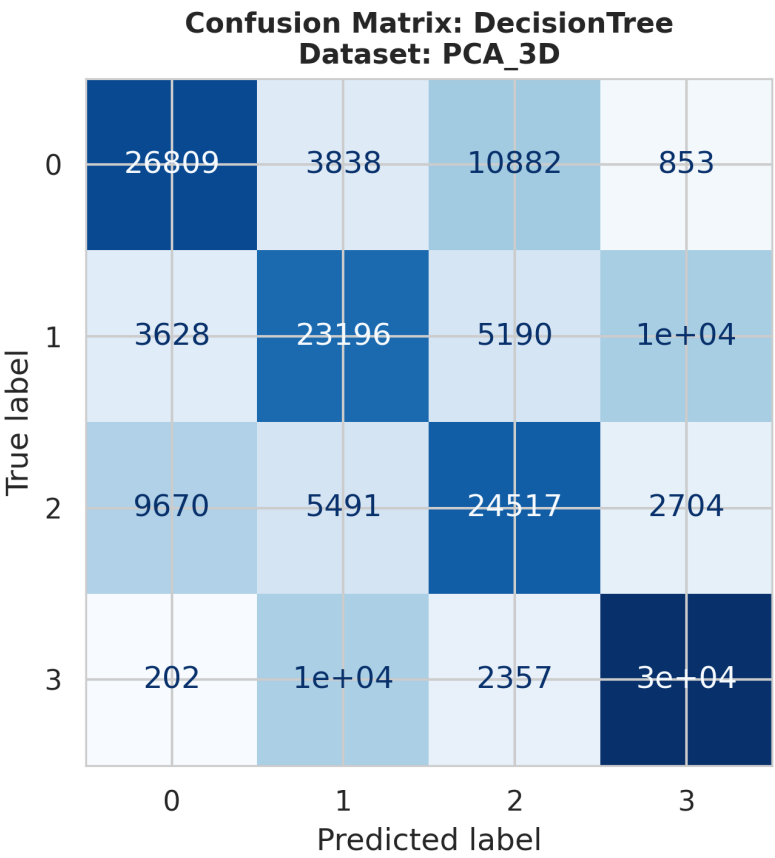
ConfusionMatrix_DecisionTree_Original_Scaled.png



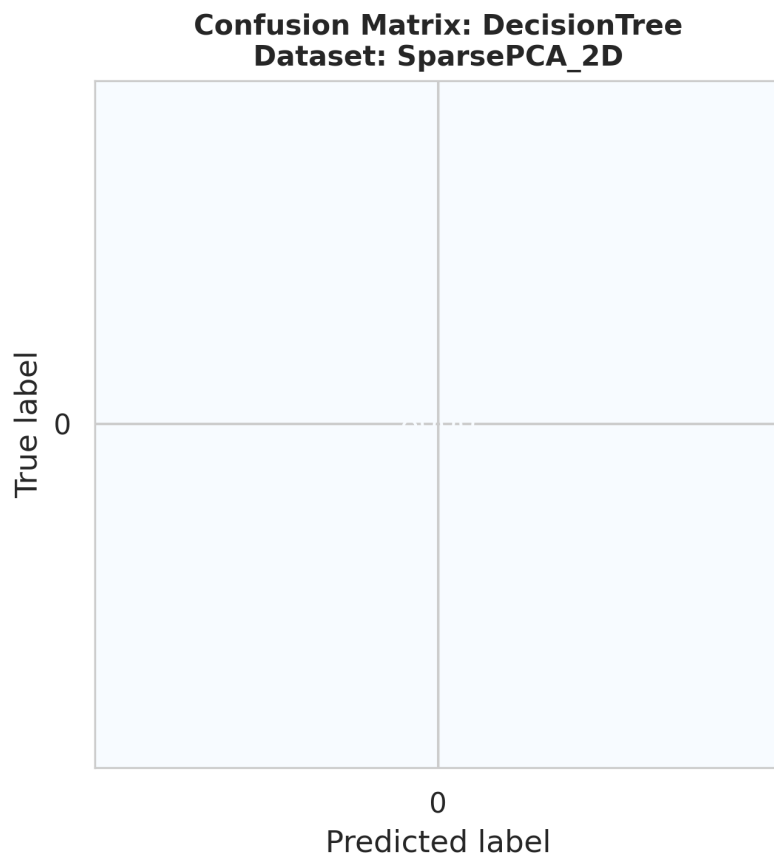
ConfusionMatrix_DecisionTree_PCA_2D.png



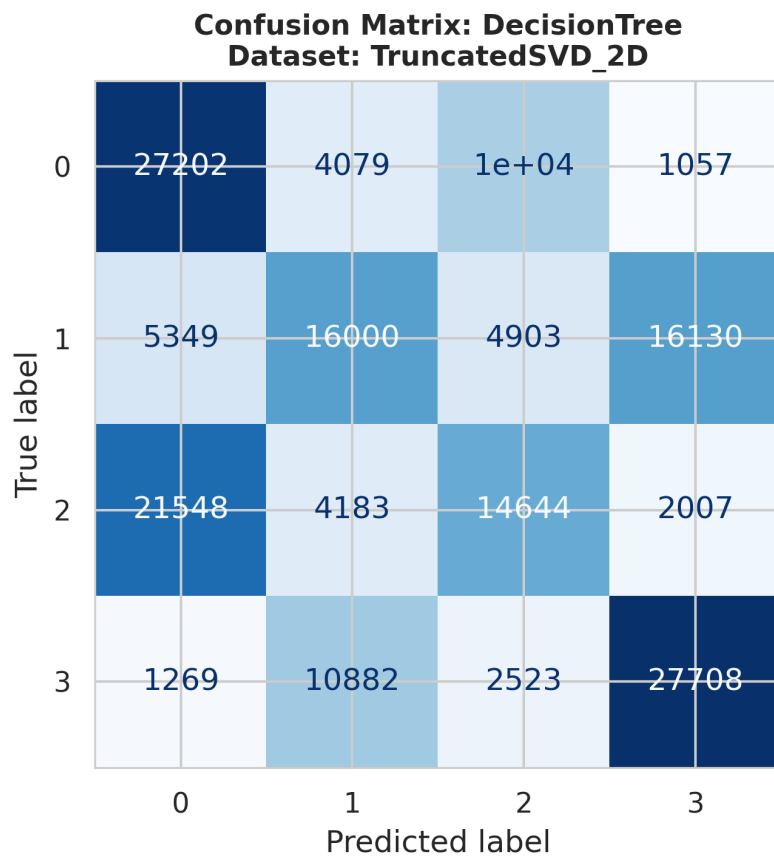
ConfusionMatrix_DecisionTree_PCA_3D.png



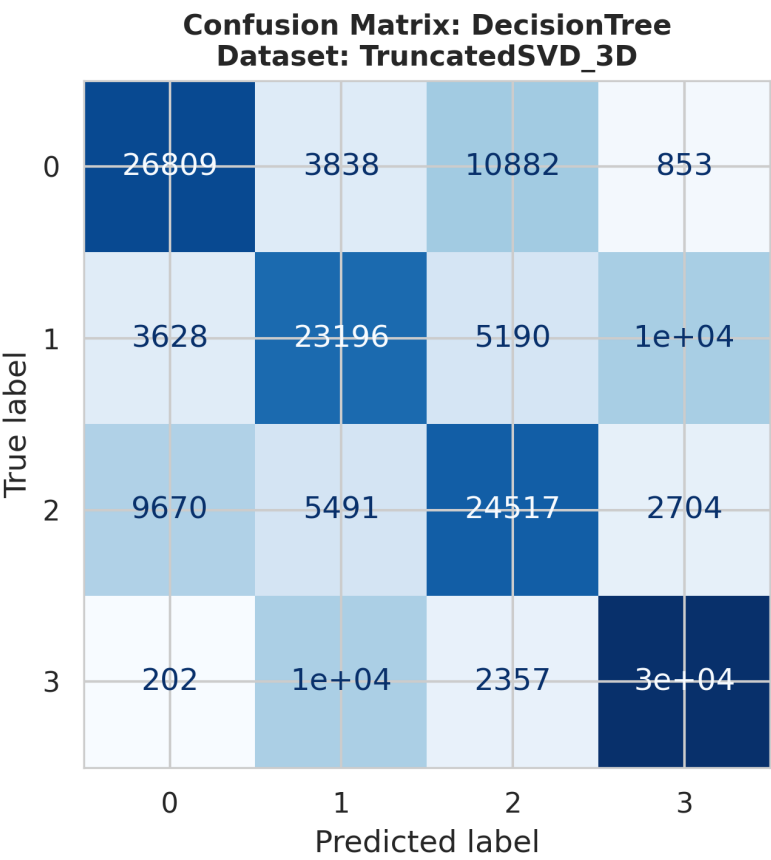
ConfusionMatrix_DecisionTree_SparsePCA_2D.png



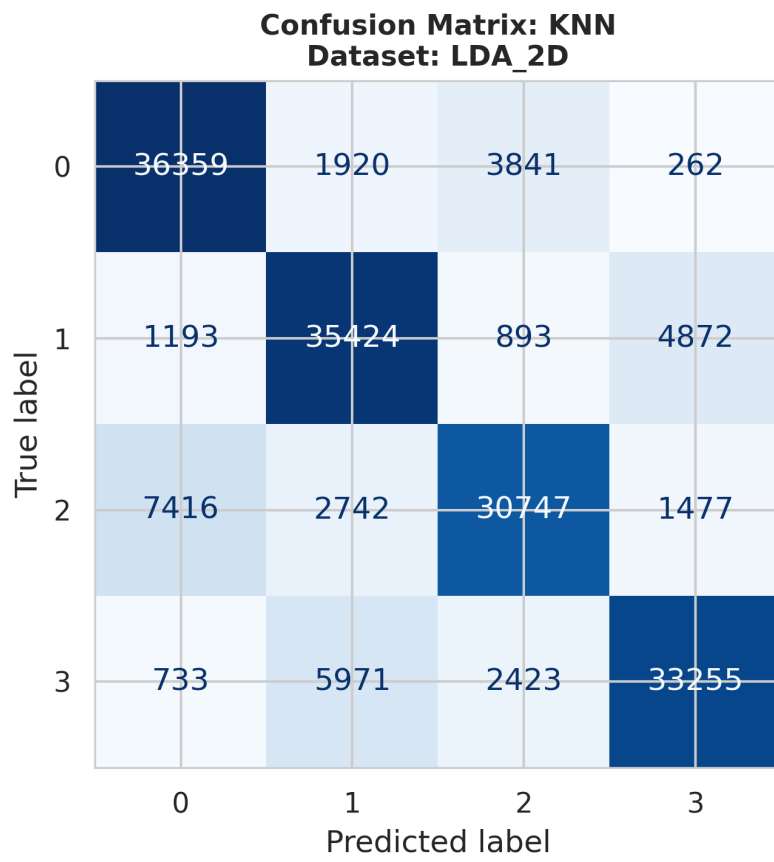
ConfusionMatrix_DecisionTree_TruncatedSVD_2D.png



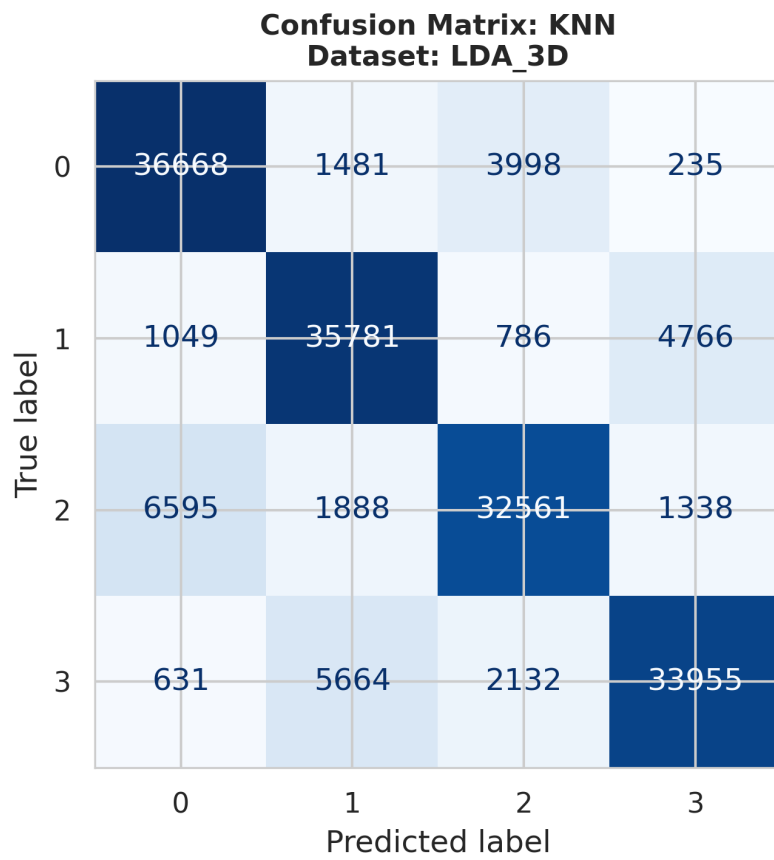
ConfusionMatrix_DecisionTree_TruncatedSVD_3D.png



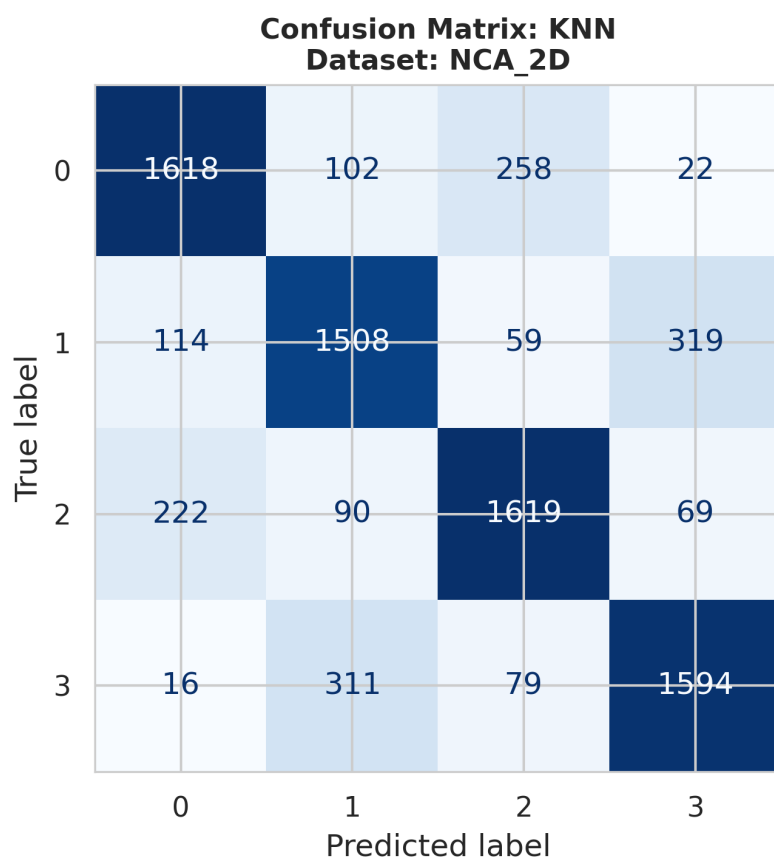
ConfusionMatrix_KNN_LDA_2D.png



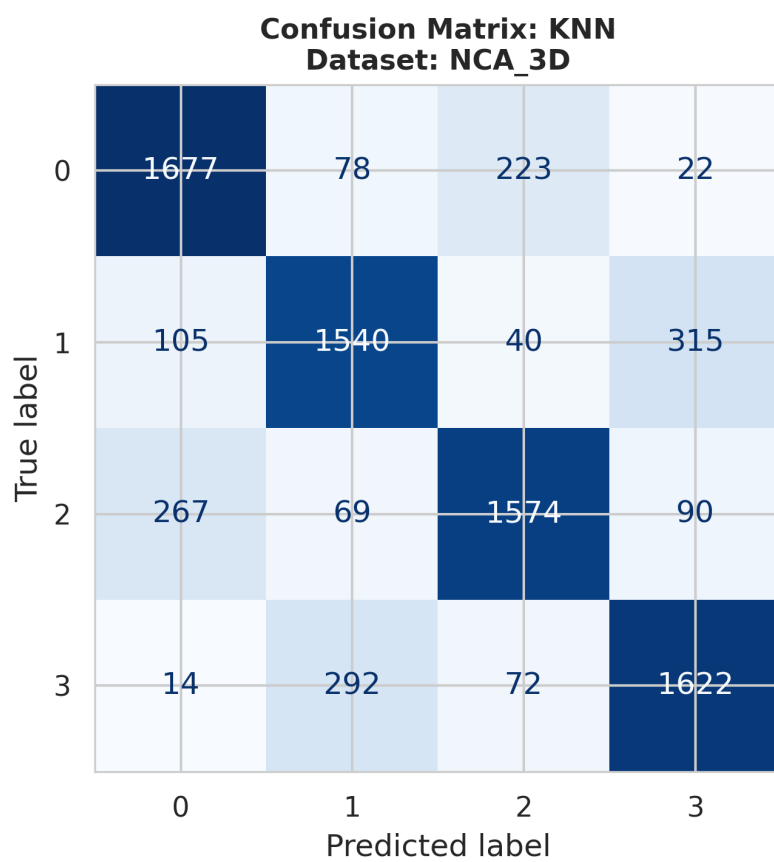
ConfusionMatrix_KNN_LDA_3D.png



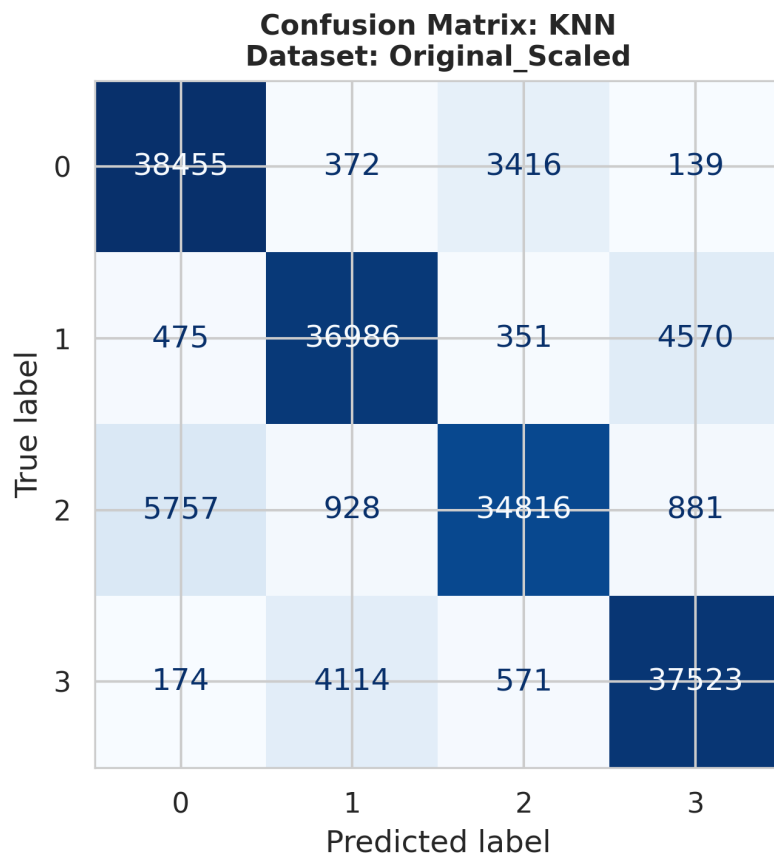
ConfusionMatrix_KNN_NCA_2D.png



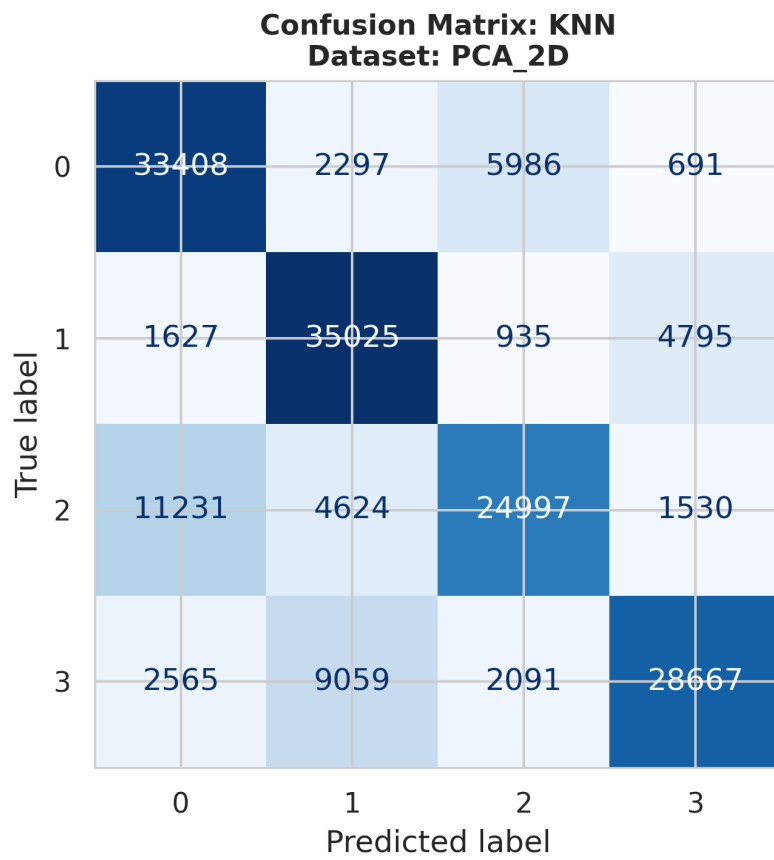
ConfusionMatrix_KNN_NCA_3D.png



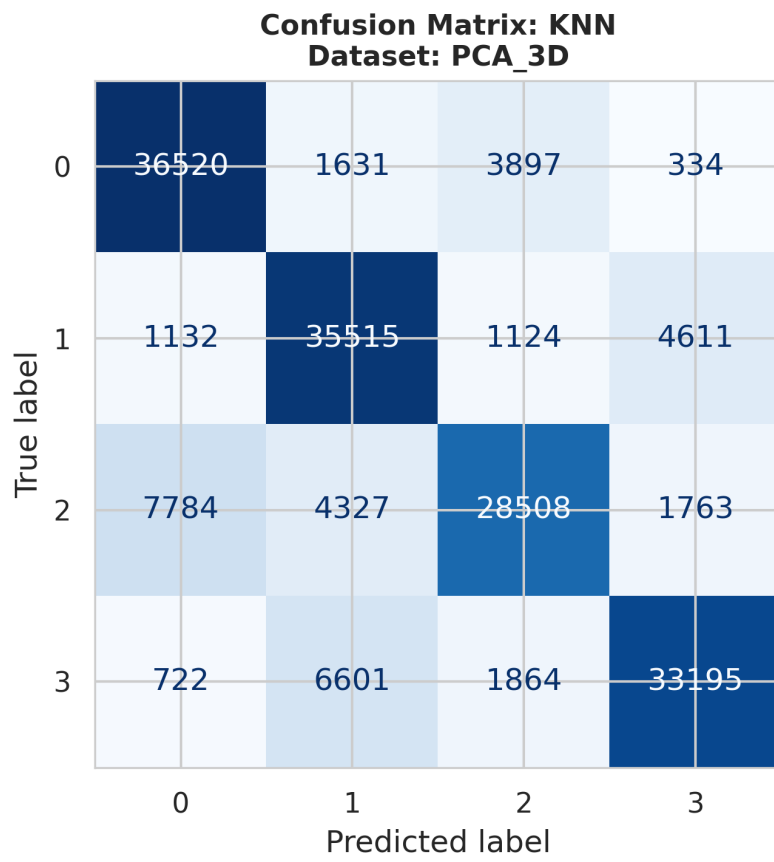
ConfusionMatrix_KNN_Original_Scaled.png



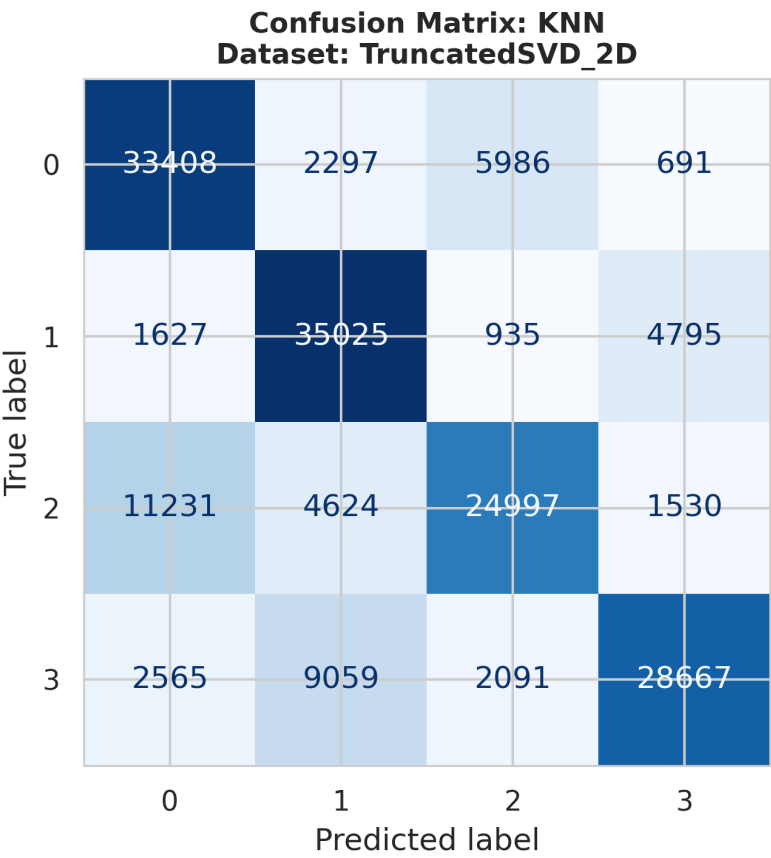
ConfusionMatrix_KNN_PCA_2D.png



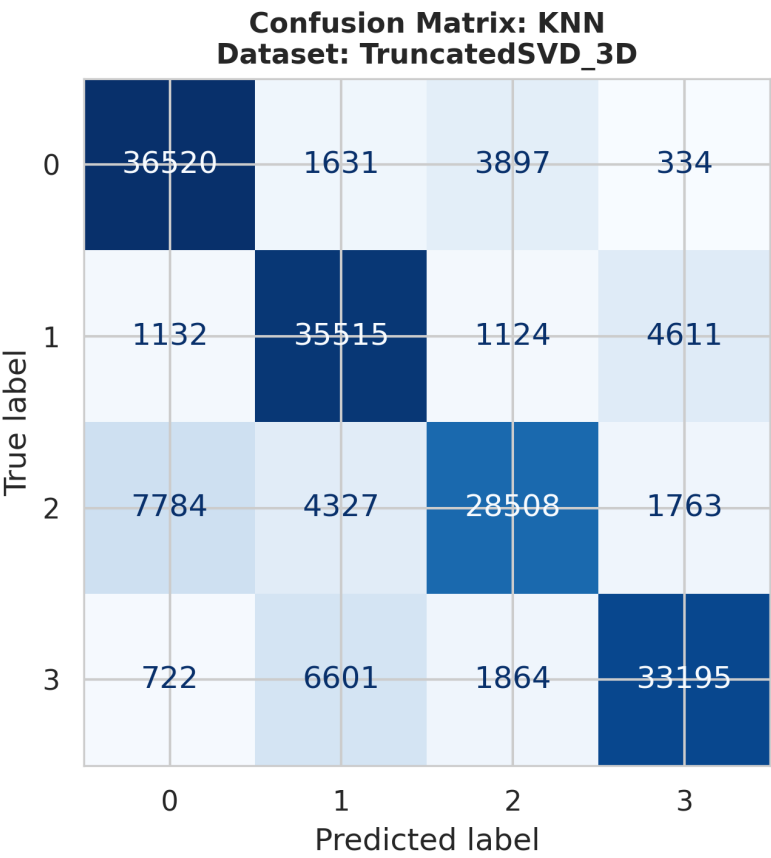
ConfusionMatrix_KNN_PCA_3D.png



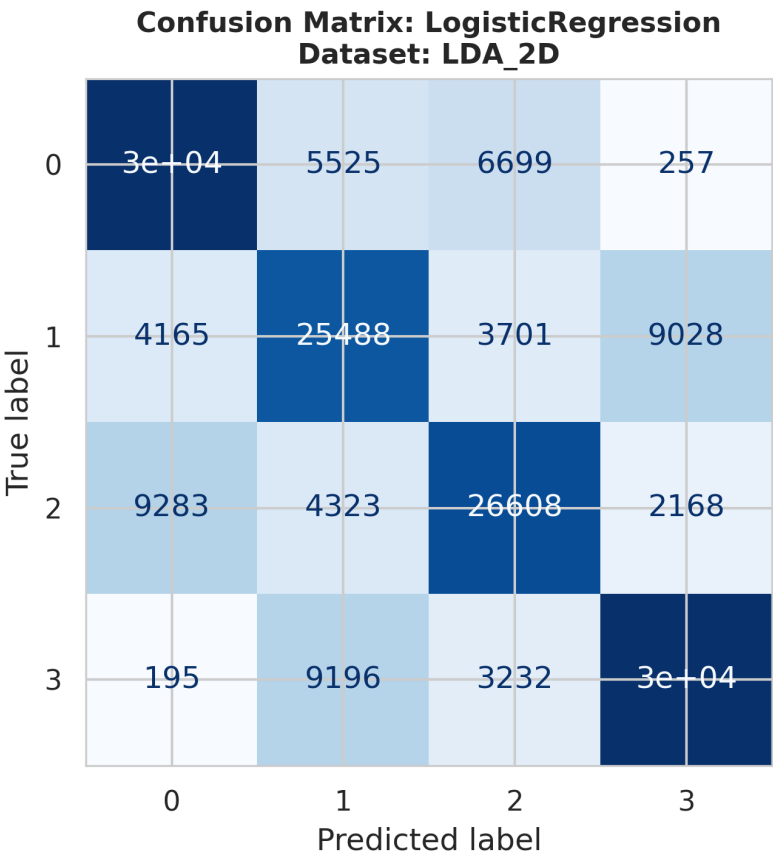
ConfusionMatrix_KNN_TruncatedSVD_2D.png



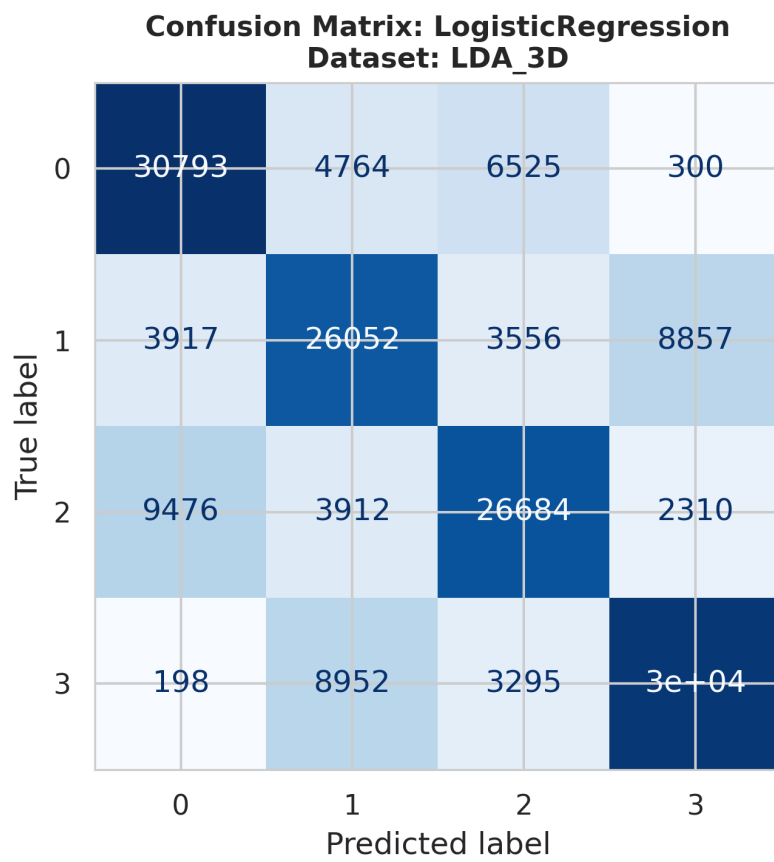
ConfusionMatrix_KNN_TruncatedSVD_3D.png



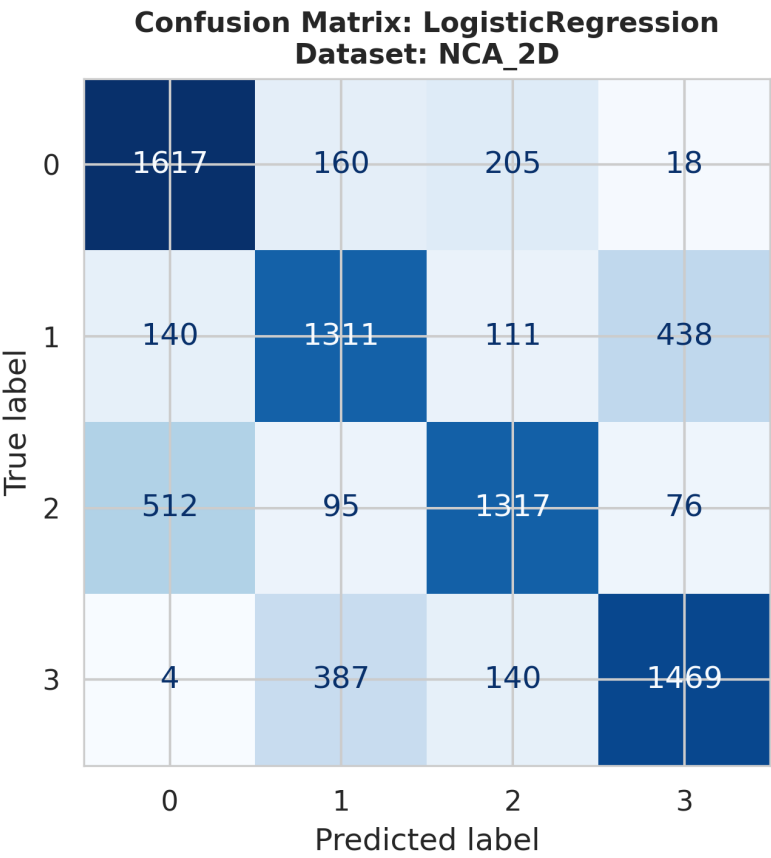
ConfusionMatrix_LogisticRegression_LDA_2D.png



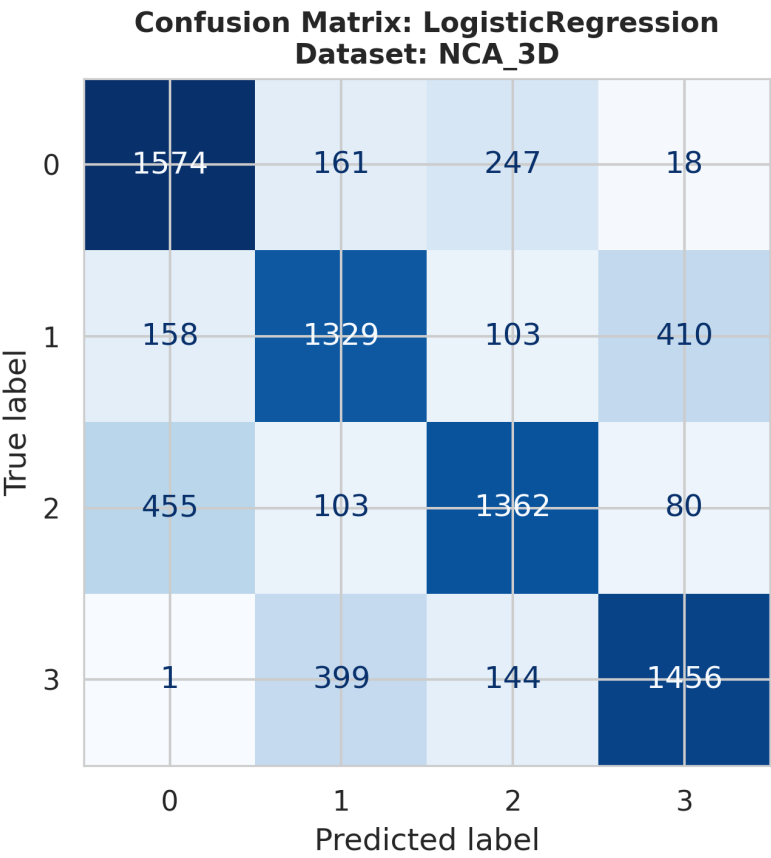
ConfusionMatrix_LogisticRegression_LDA_3D.png



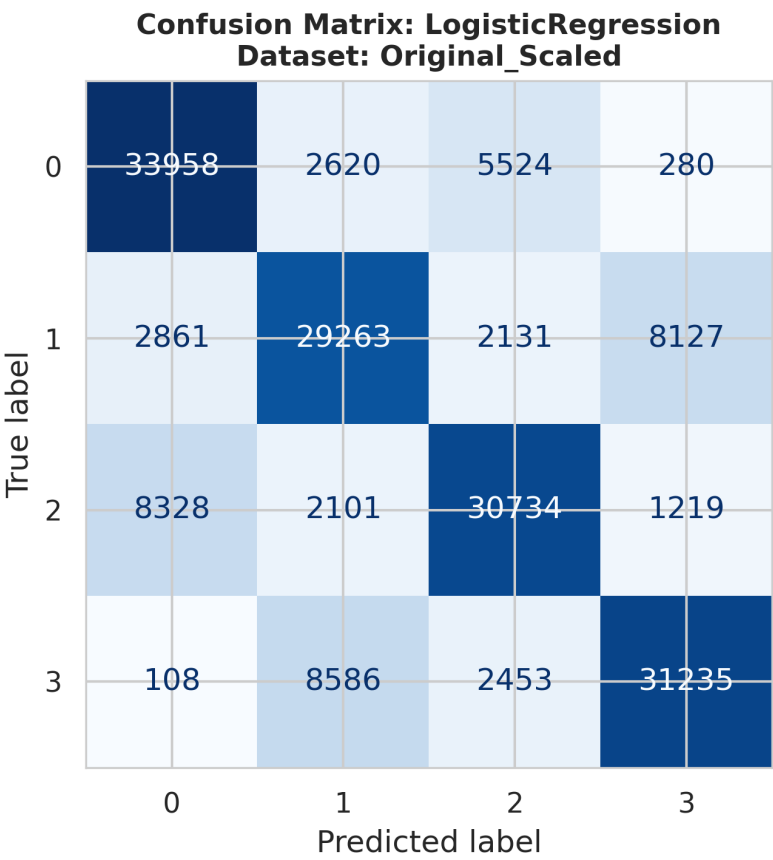
ConfusionMatrix_LogisticRegression_NCA_2D.png



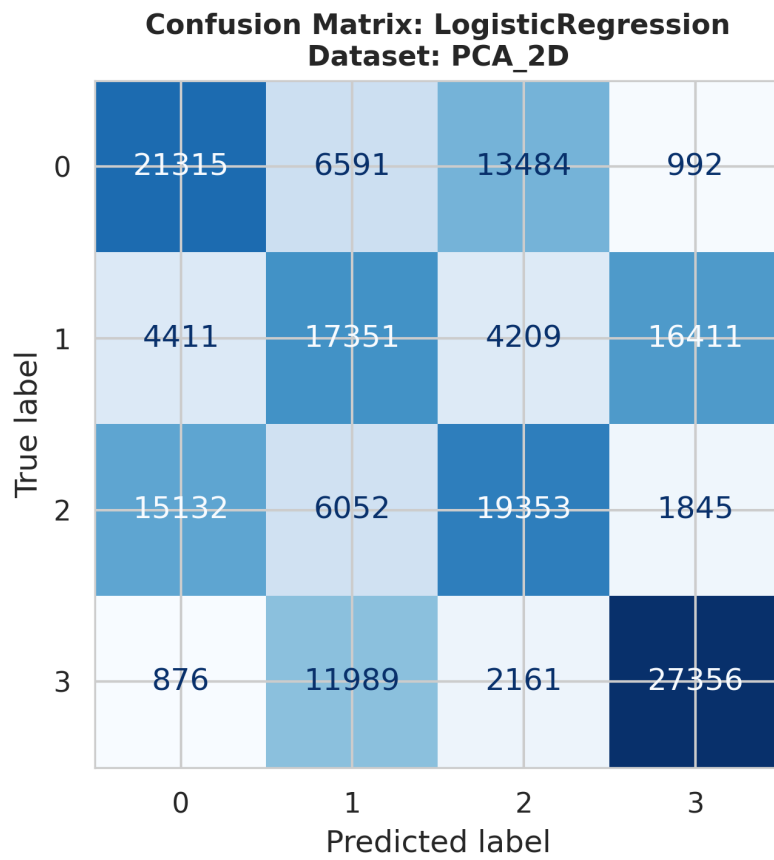
ConfusionMatrix_LogisticRegression_NCA_3D.png



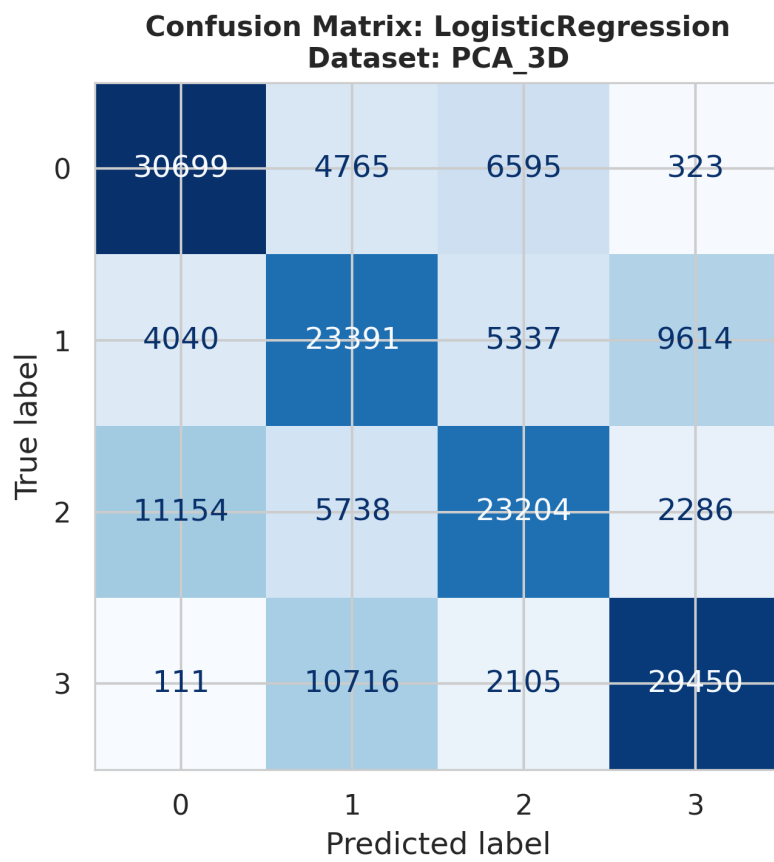
ConfusionMatrix_LogisticRegression_Original_Scaled.png



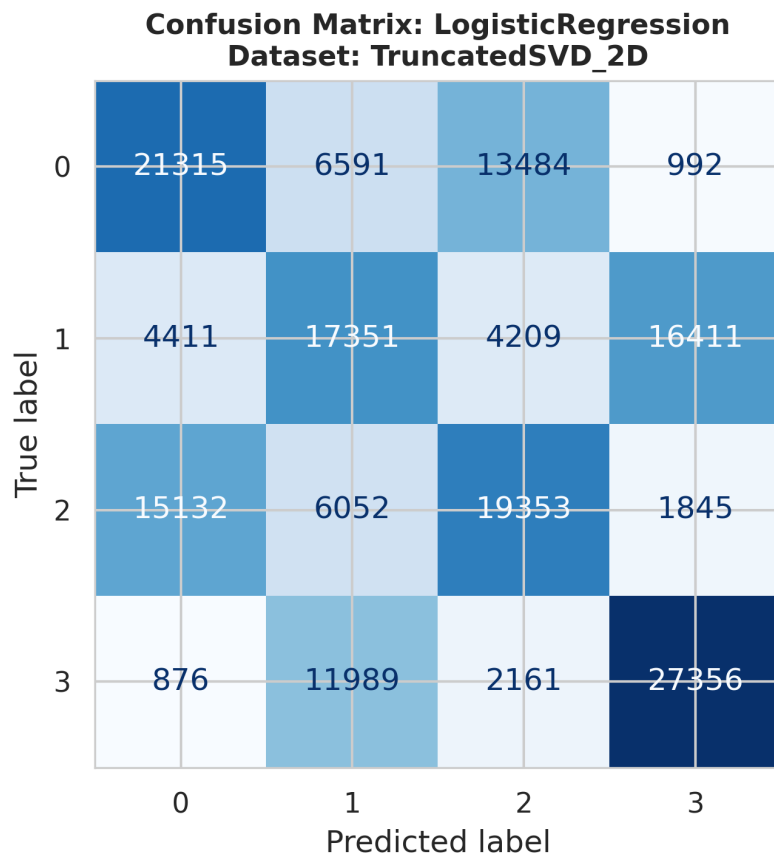
ConfusionMatrix_LogisticRegression_PCA_2D.png



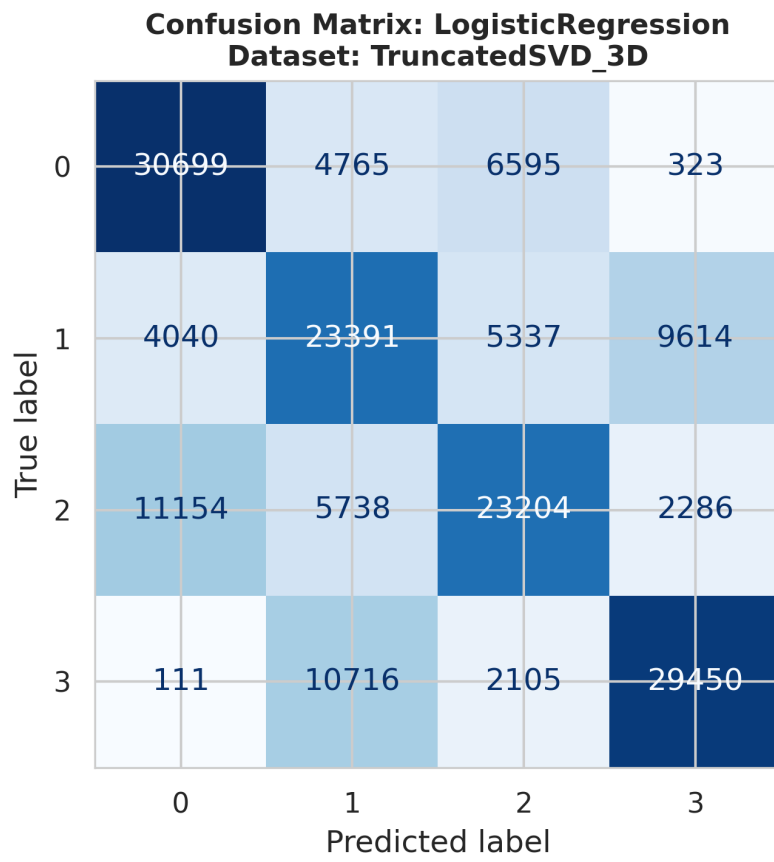
ConfusionMatrix_LogisticRegression_PCA_3D.png



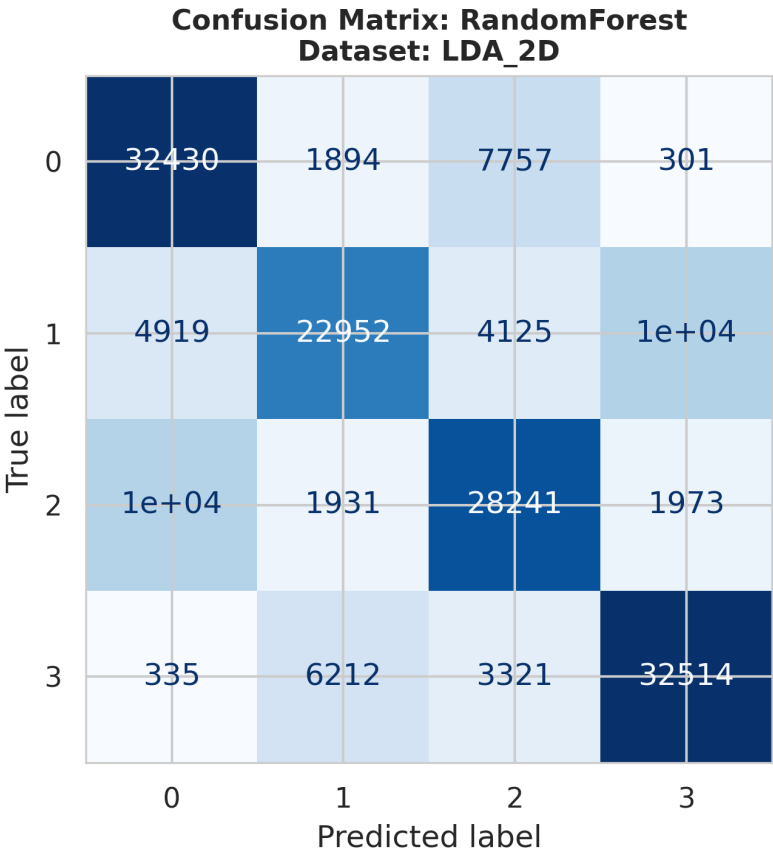
ConfusionMatrix_LogisticRegression_TruncatedSVD_2D.png



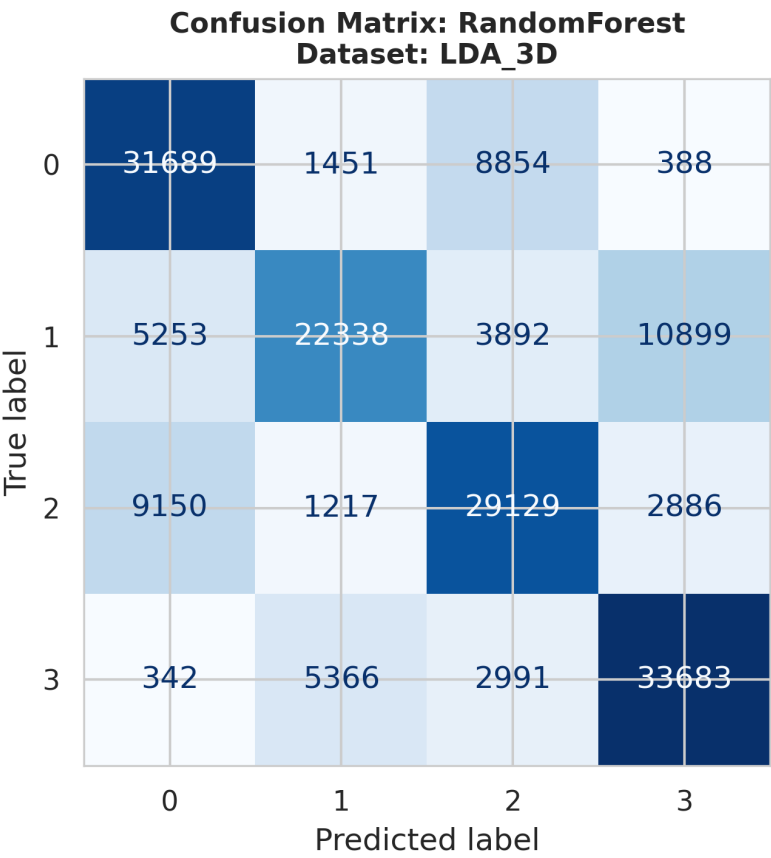
ConfusionMatrix_LogisticRegression_TruncatedSVD_3D.png



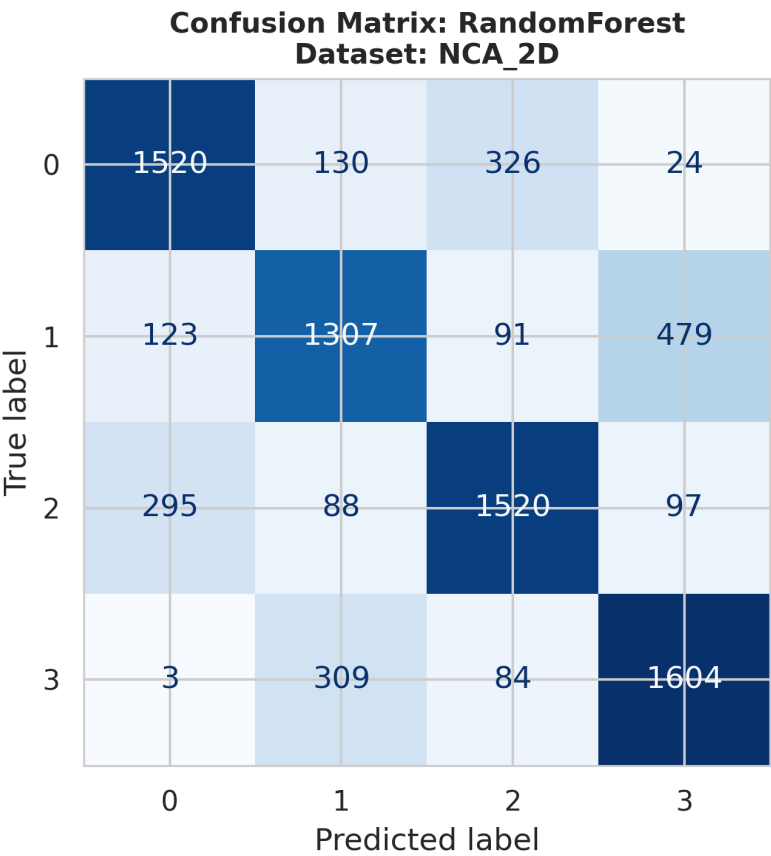
ConfusionMatrix_RandomForest_LDA_2D.png



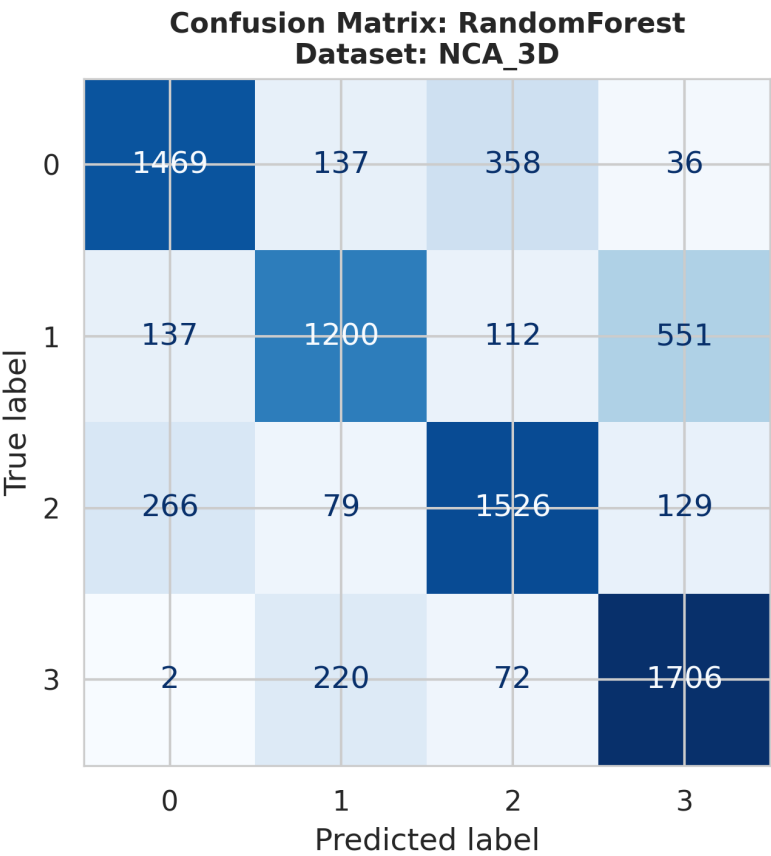
ConfusionMatrix_RandomForest_LDA_3D.png



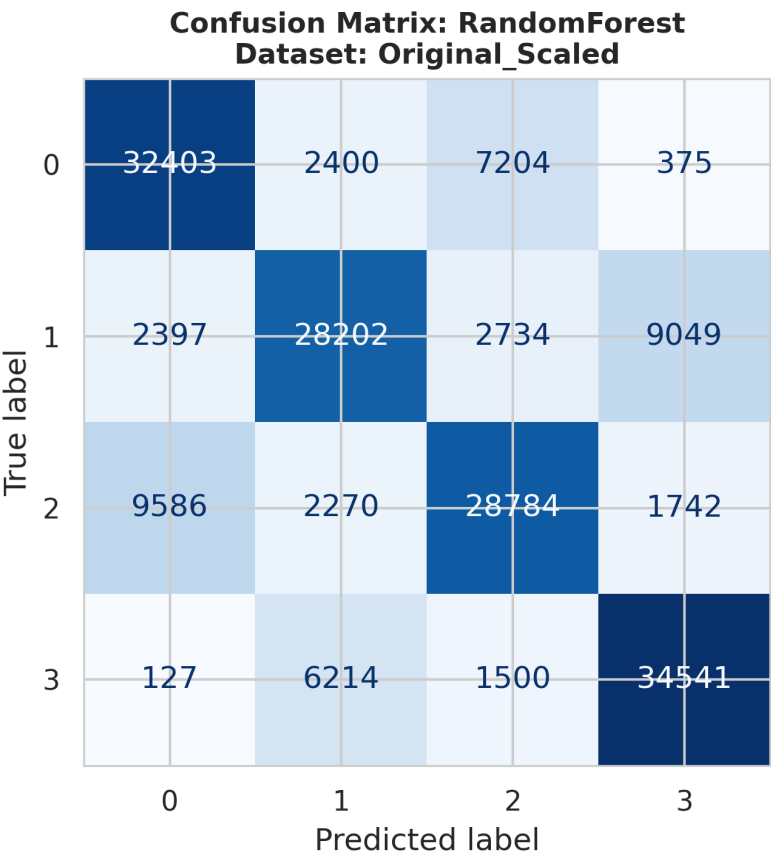
ConfusionMatrix_RandomForest_NCA_2D.png



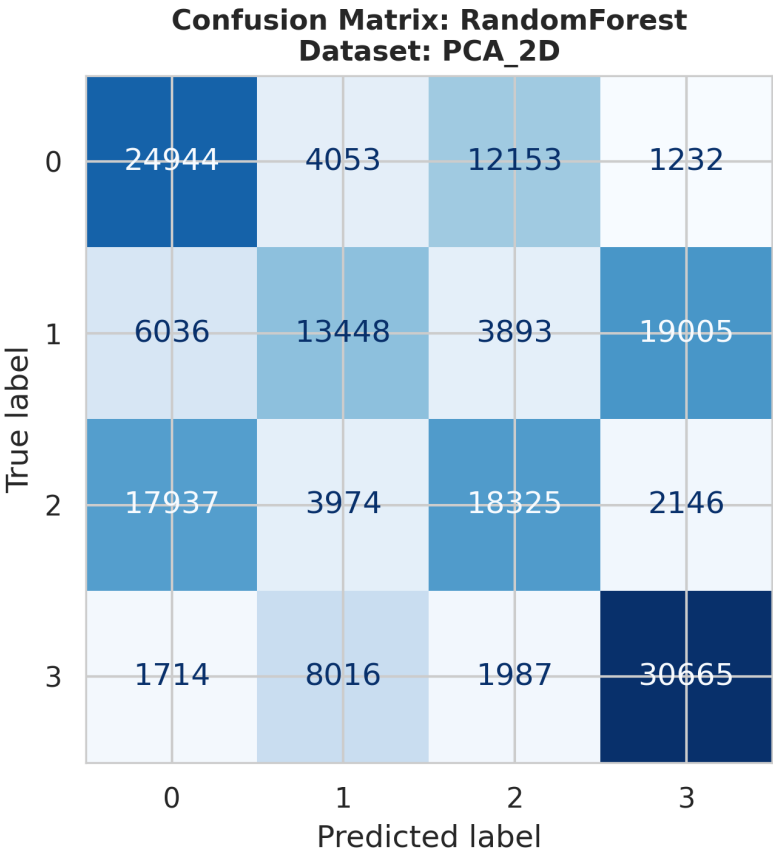
ConfusionMatrix_RandomForest_NCA_3D.png



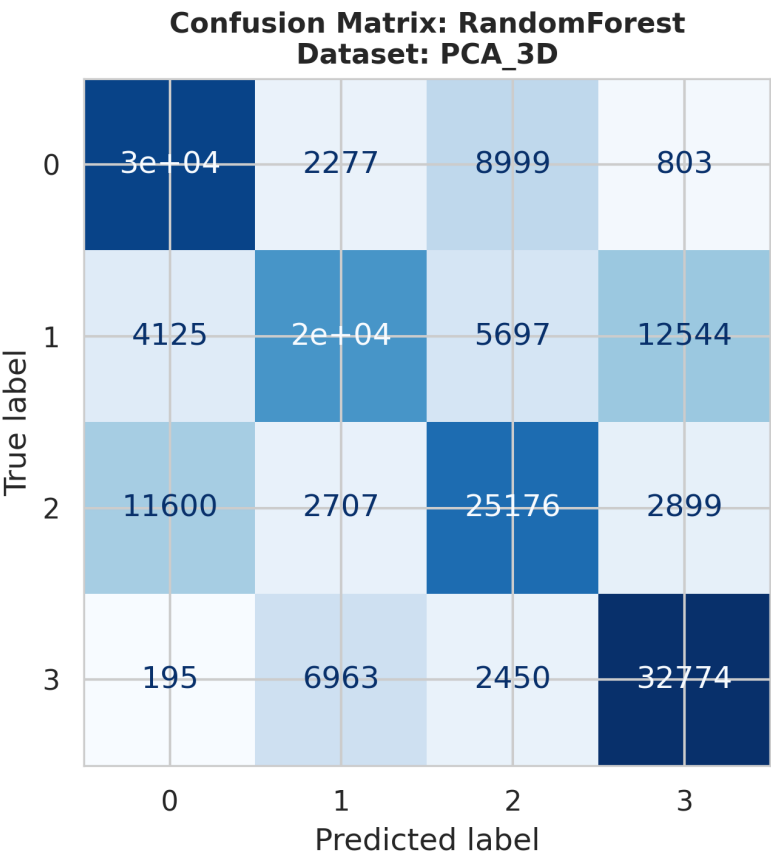
ConfusionMatrix_RandomForest_Original_Scaled.png



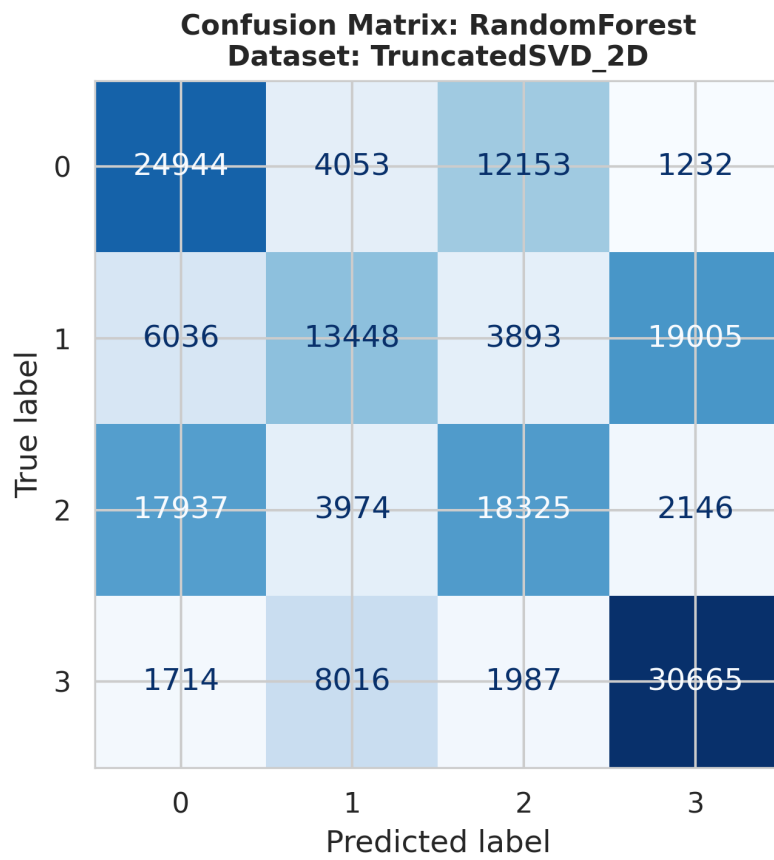
ConfusionMatrix_RandomForest_PCA_2D.png



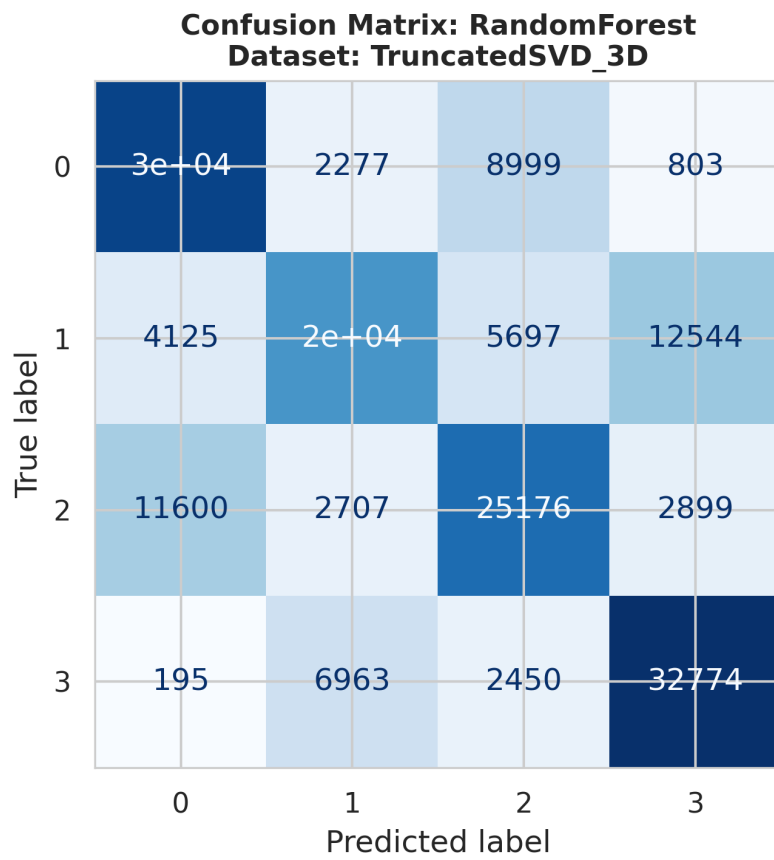
ConfusionMatrix_RandomForest_PCA_3D.png



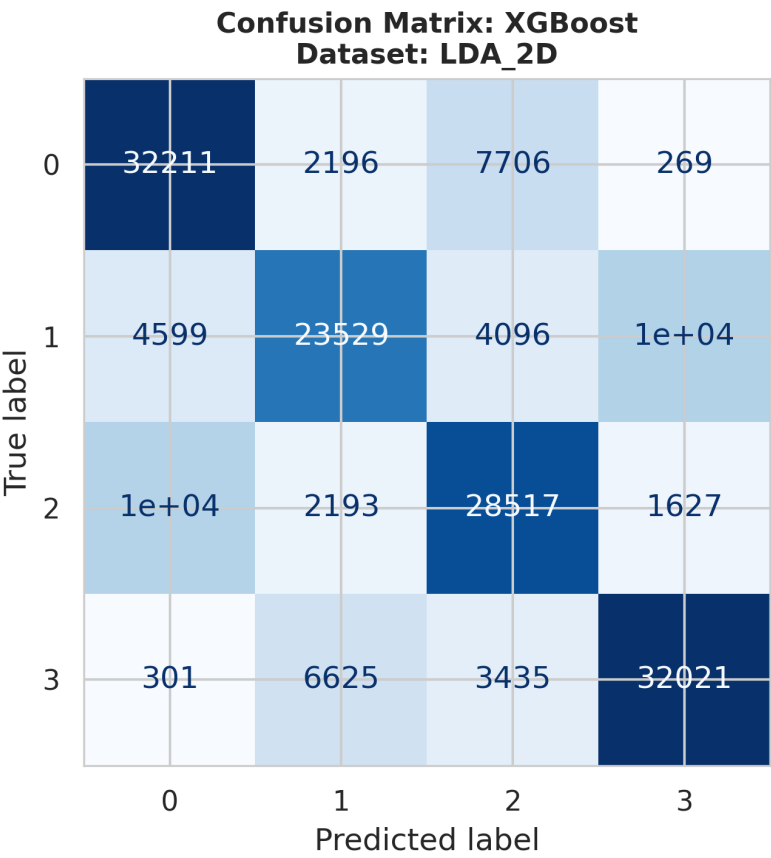
ConfusionMatrix_RandomForest_TruncatedSVD_2D.png



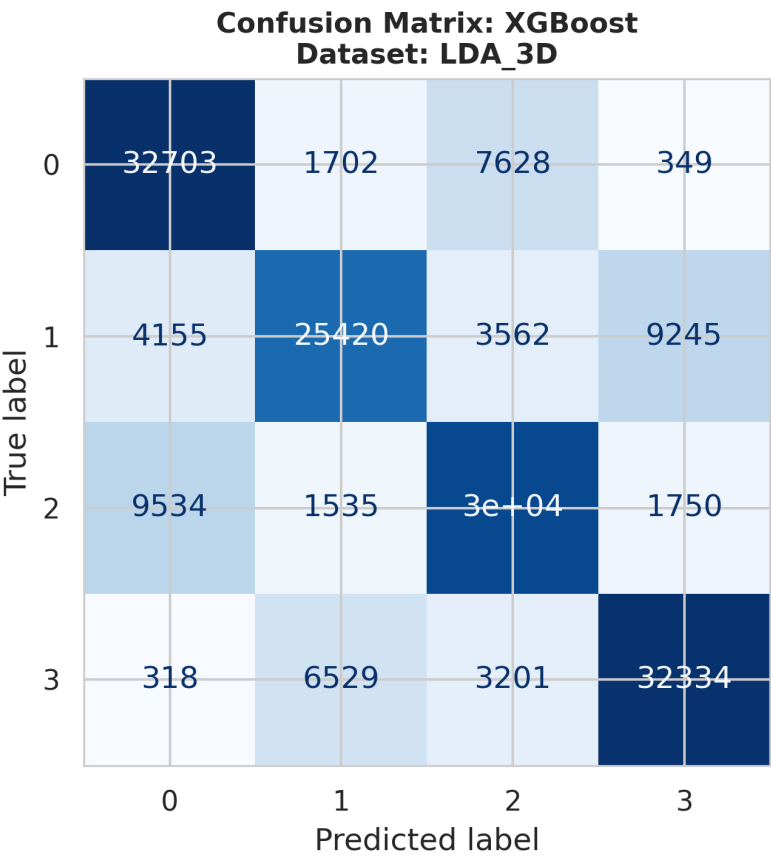
ConfusionMatrix_RandomForest_TruncatedSVD_3D.png



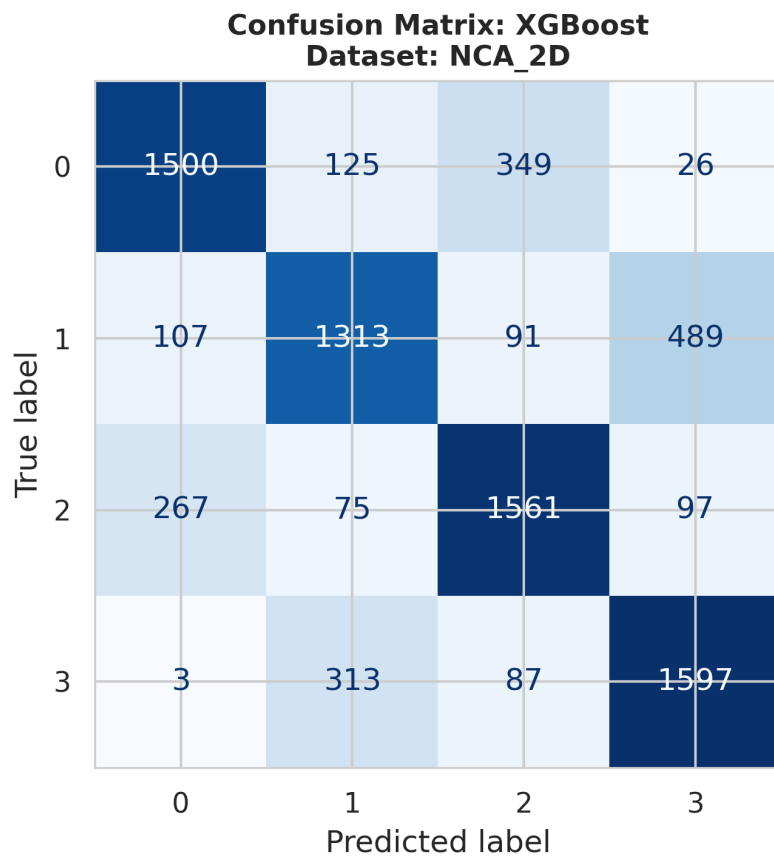
ConfusionMatrix_XGBoost_LDA_2D.png



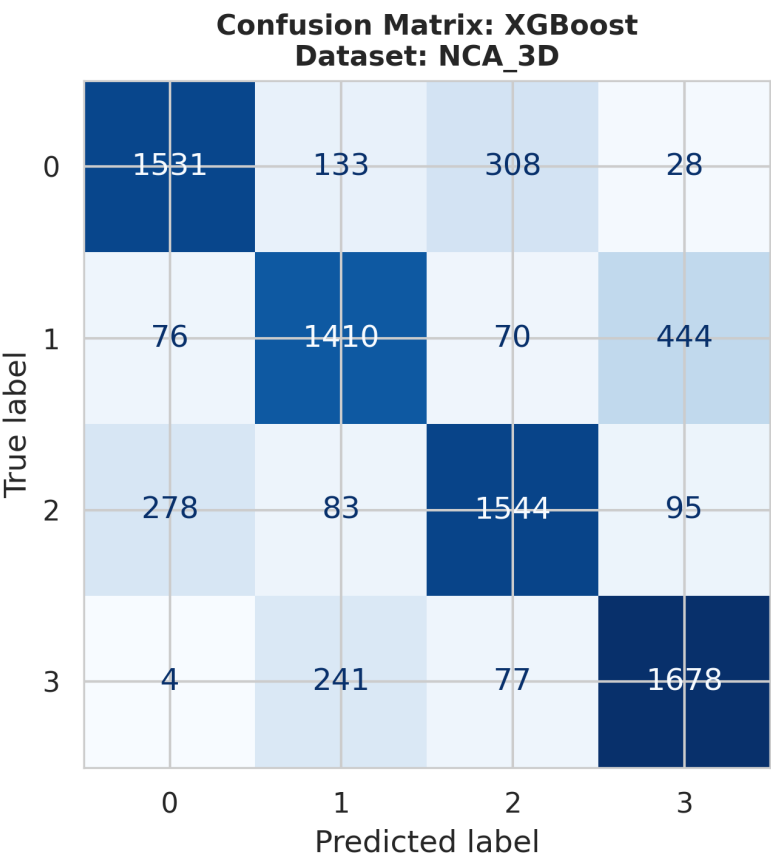
ConfusionMatrix_XGBoost_LDA_3D.png



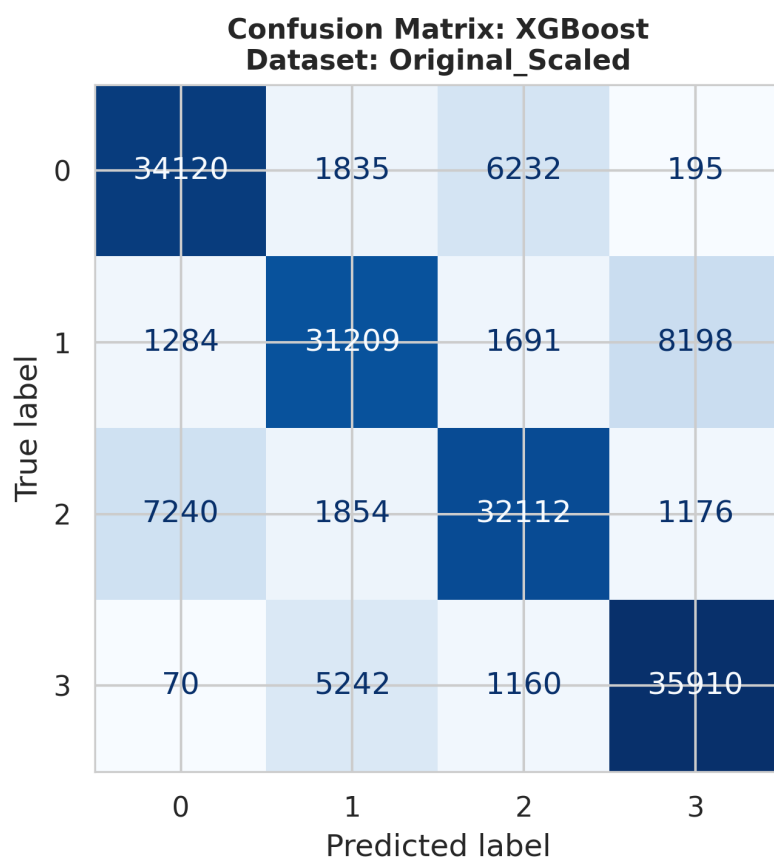
ConfusionMatrix_XGBoost_NCA_2D.png



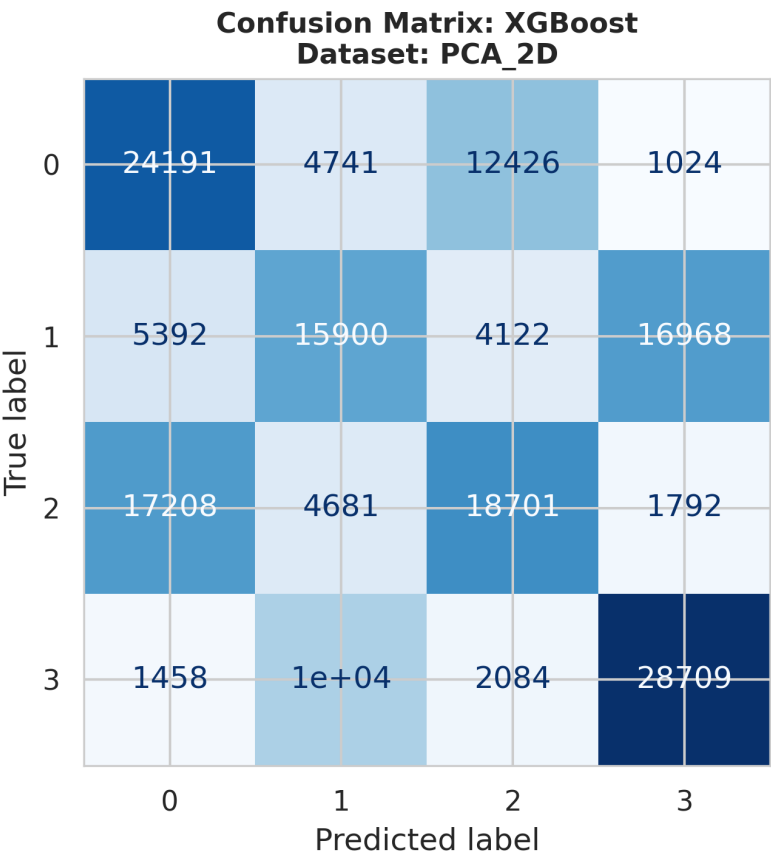
ConfusionMatrix_XGBoost_NCA_3D.png



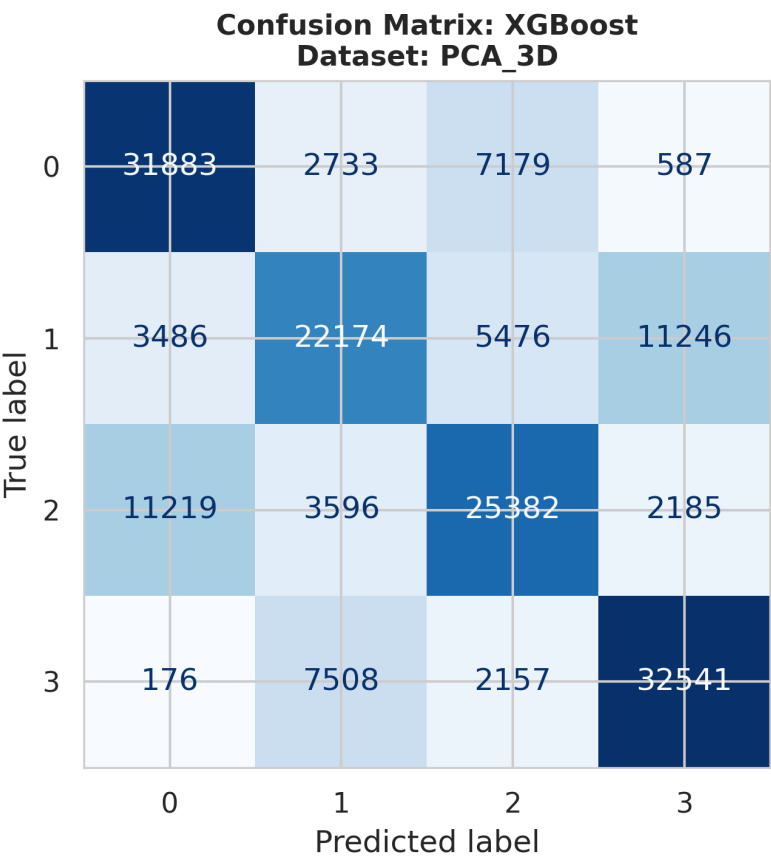
ConfusionMatrix_XGBoost_Original_Scaled.png



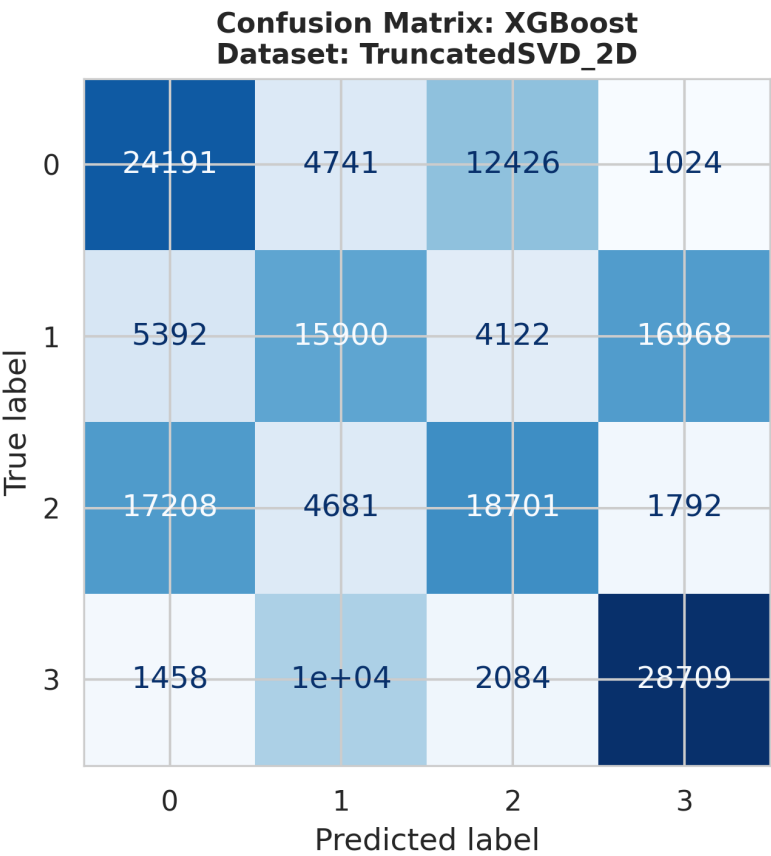
ConfusionMatrix_XGBoost_PCA_2D.png



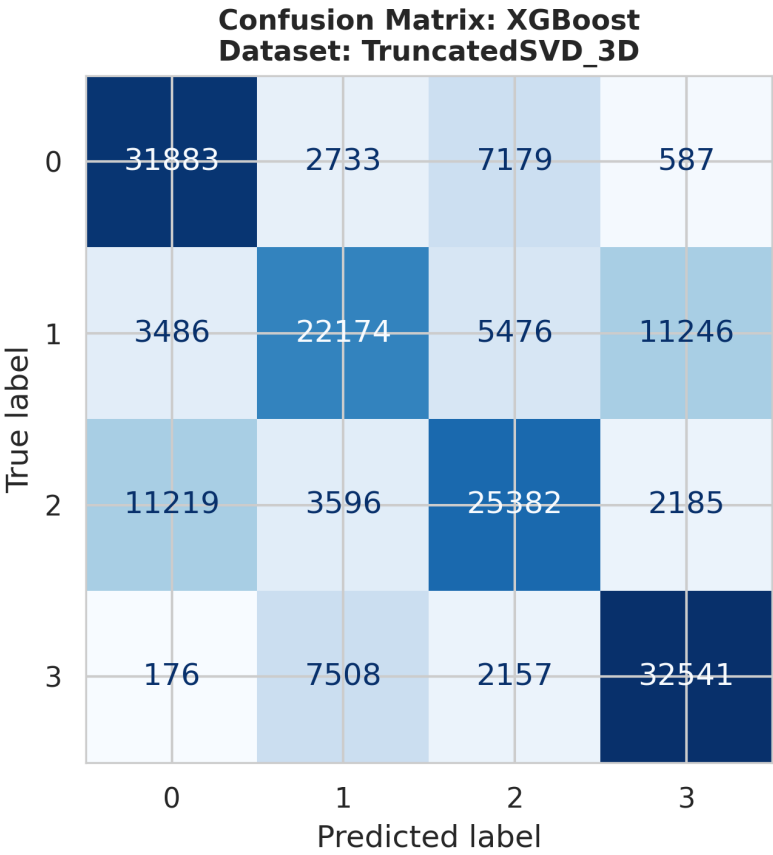
ConfusionMatrix_XGBoost_PCA_3D.png



ConfusionMatrix_XGBoost_TruncatedSVD_2D.png

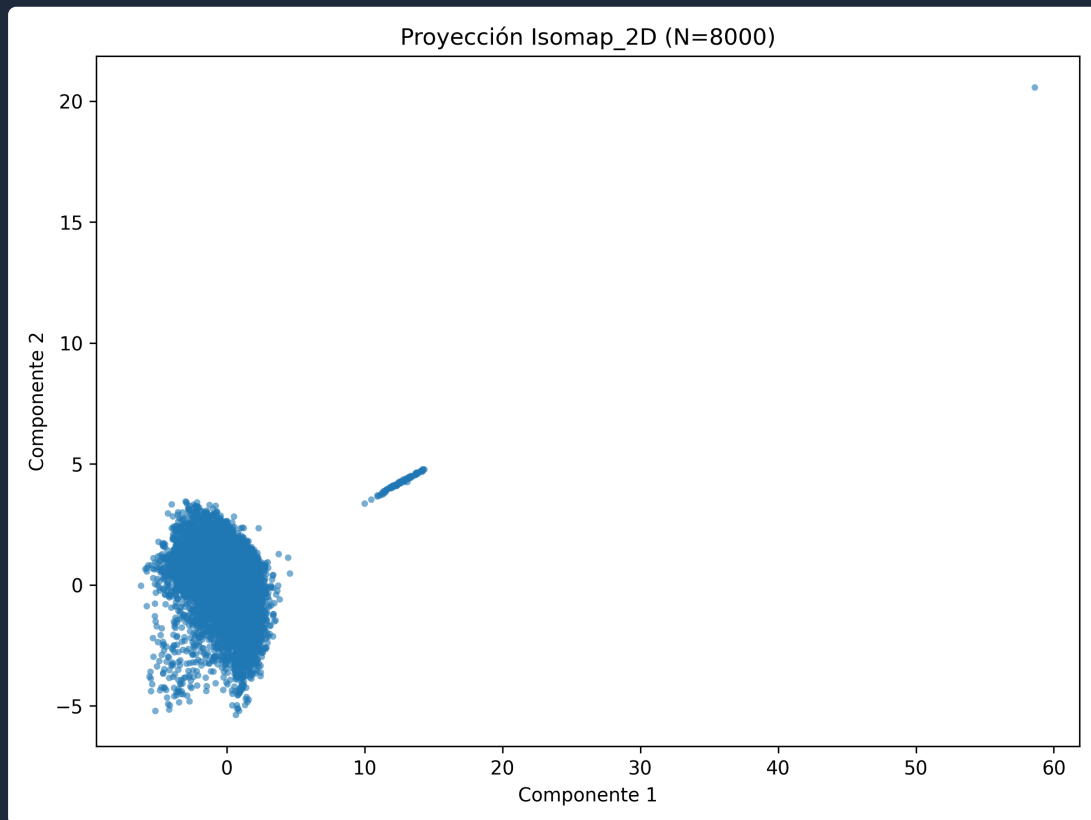


ConfusionMatrix_XGBoost_TruncatedSVD_3D.png



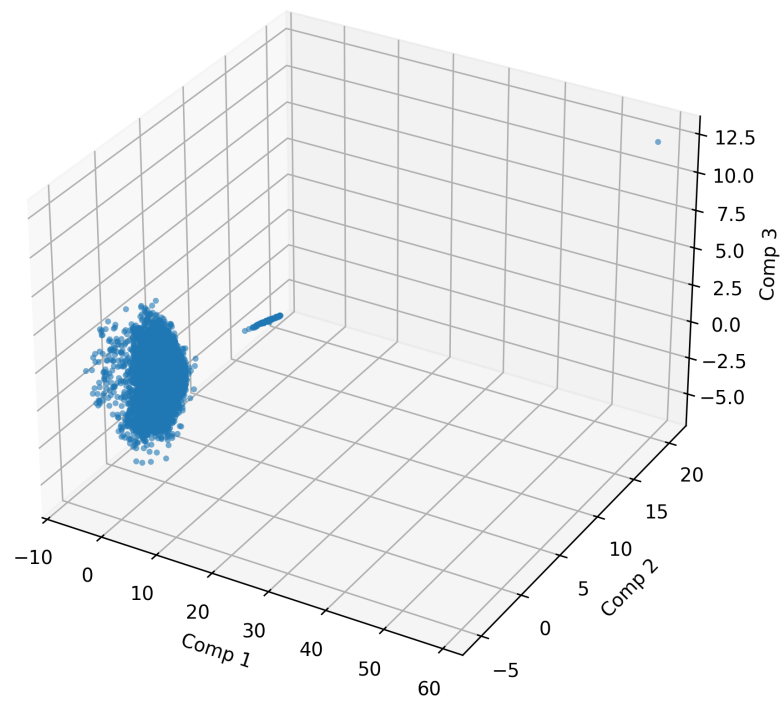
5. Proyecciones Espaciales de Reducción Dimensional (.img)

plot_Isomap_2D.png

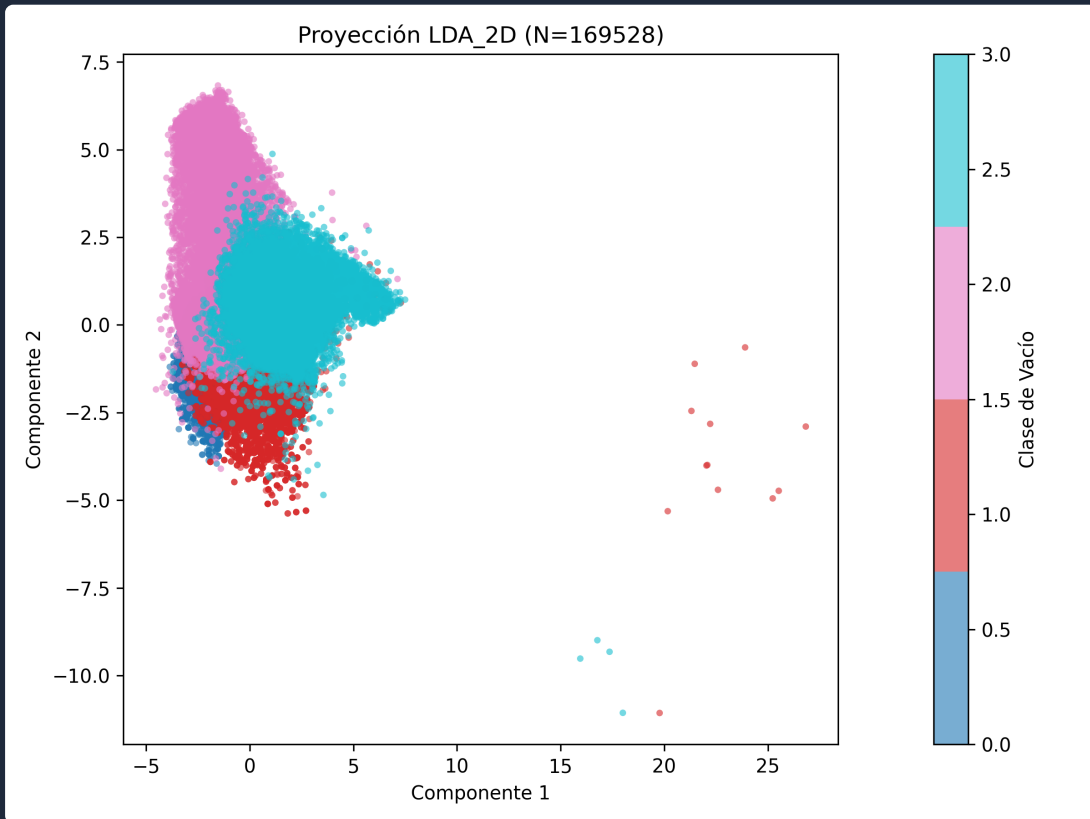


plot_Isomap_3D.png

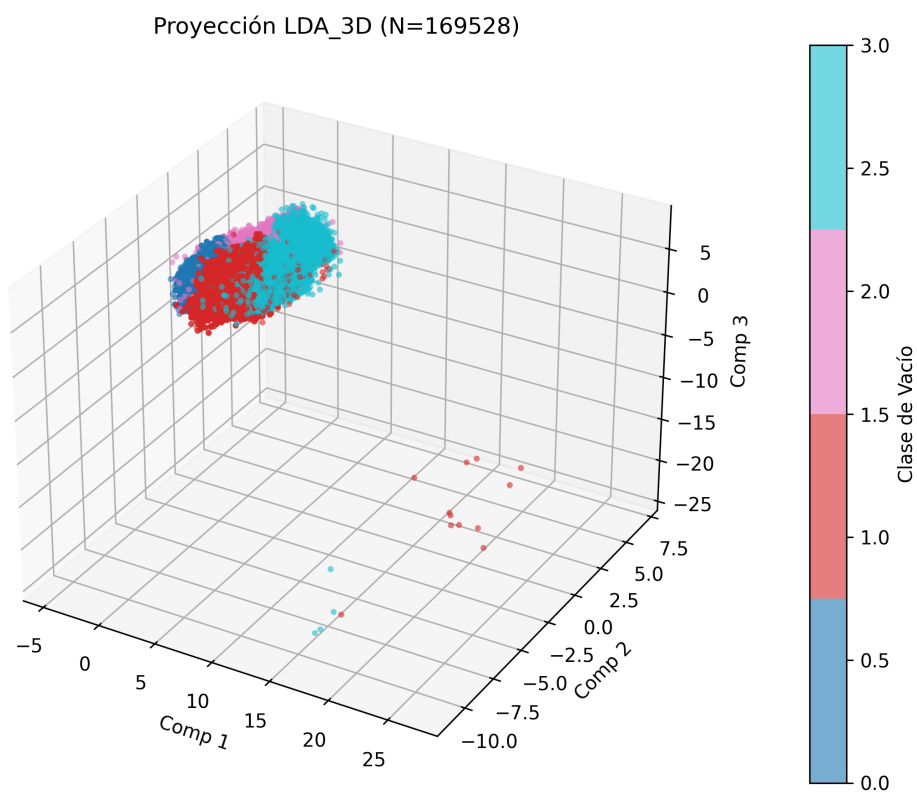
Proyección Isomap_3D (N=8000)



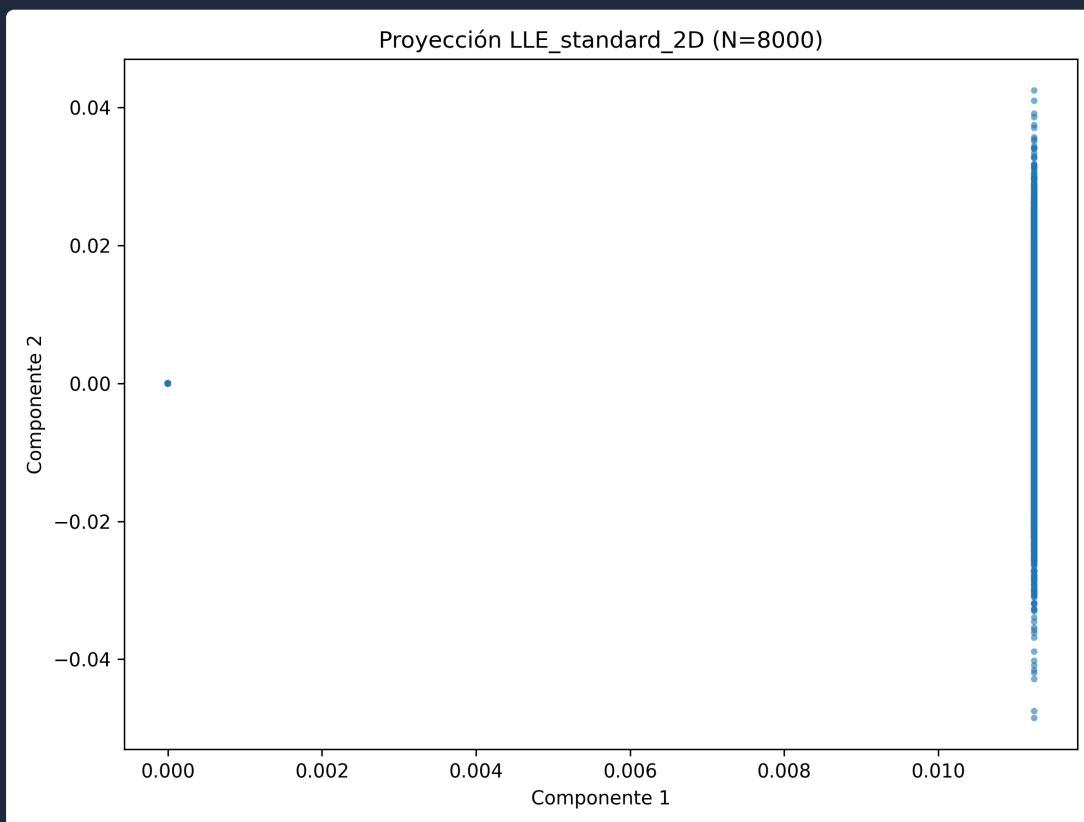
plot_LDA_2D.png



plot_LDA_3D.png

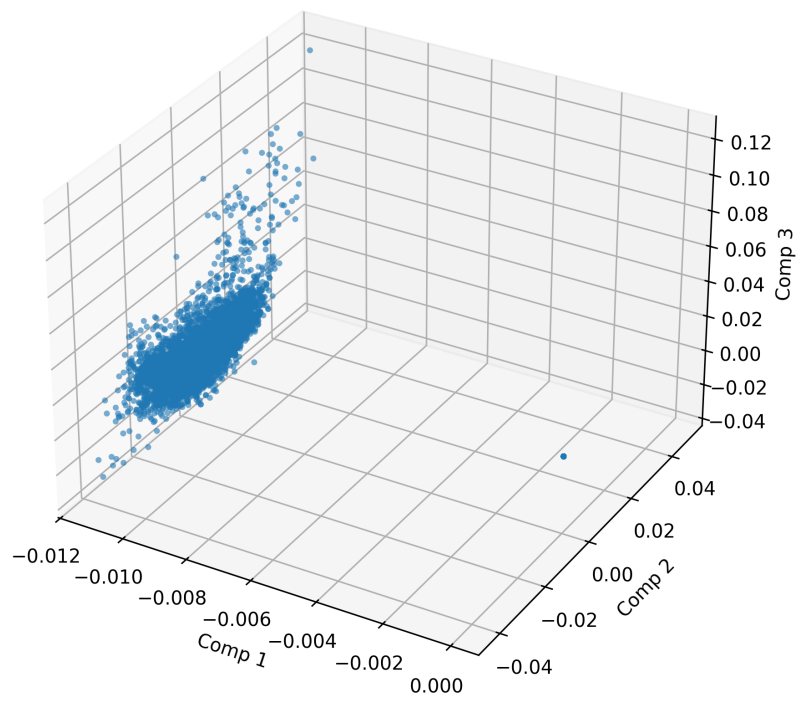


plot_LLE_standard_2D.png

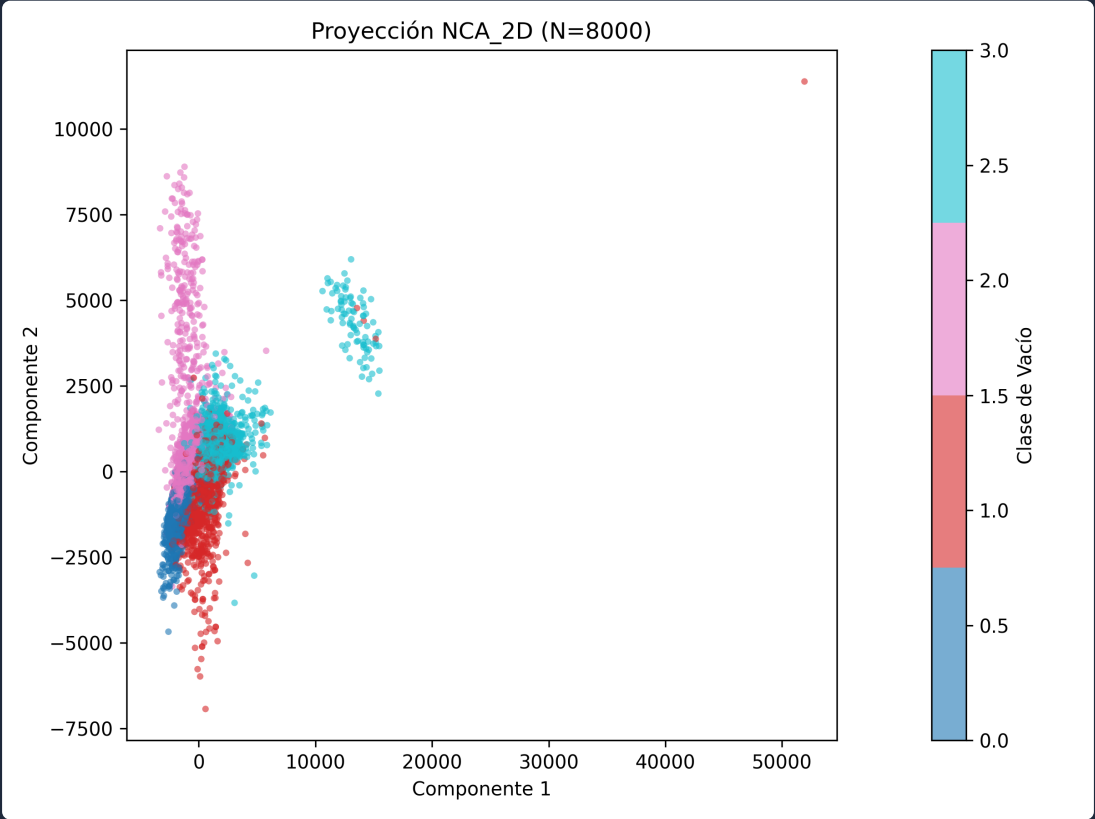


plot_LLE_standard_3D.png

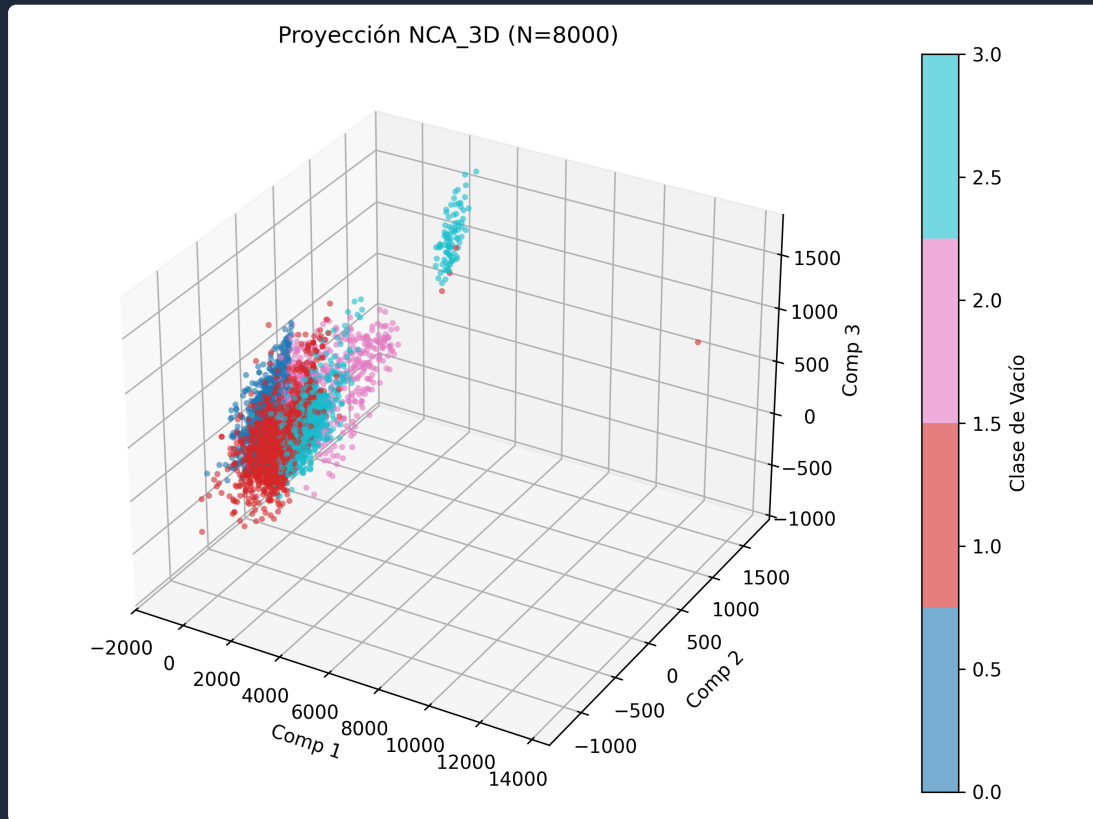
Proyección LLE_standard_3D (N=8000)



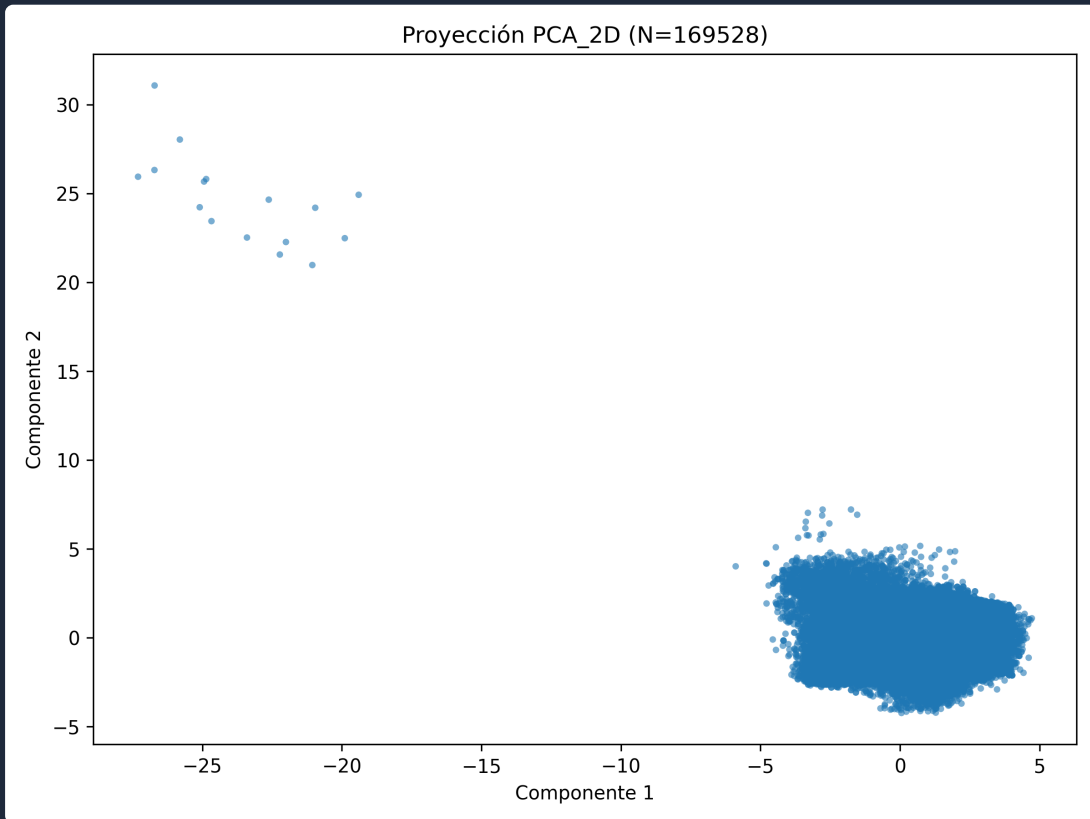
plot_NCA_2D.png



plot_NCA_3D.png

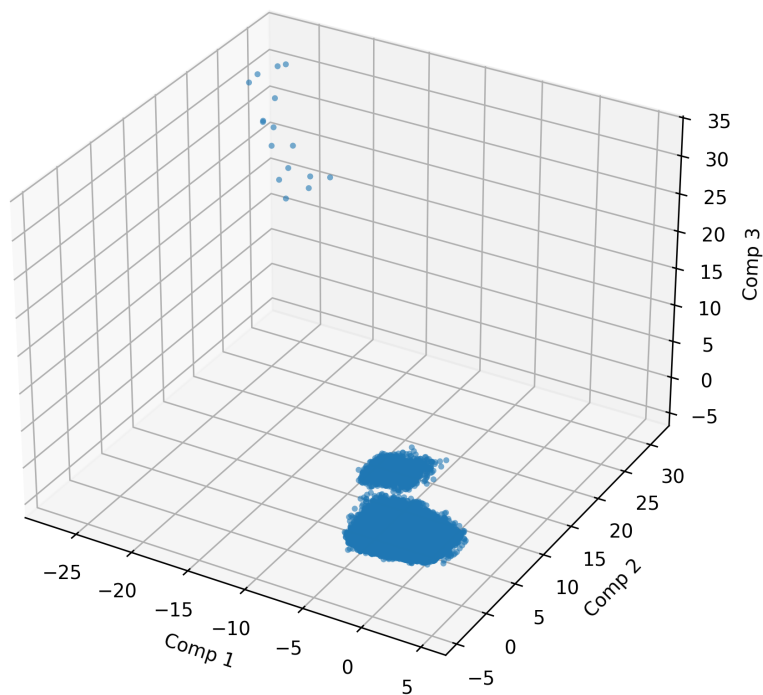


plot_PCA_2D.png

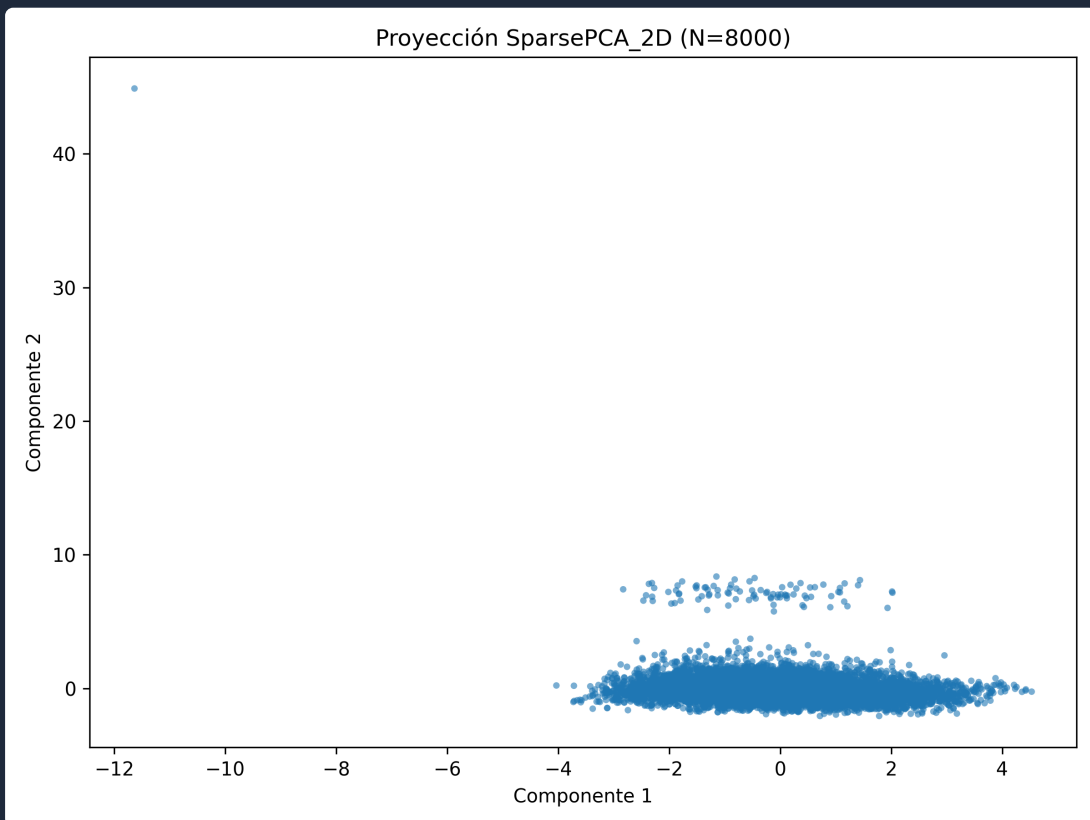


plot_PCA_3D.png

Proyección PCA_3D (N=169528)

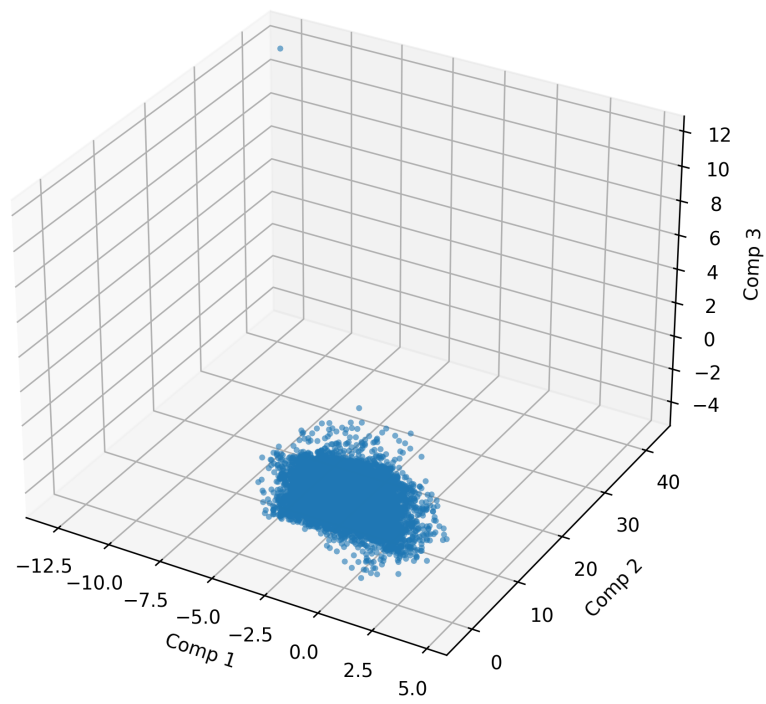


plot_SparsePCA_2D.png

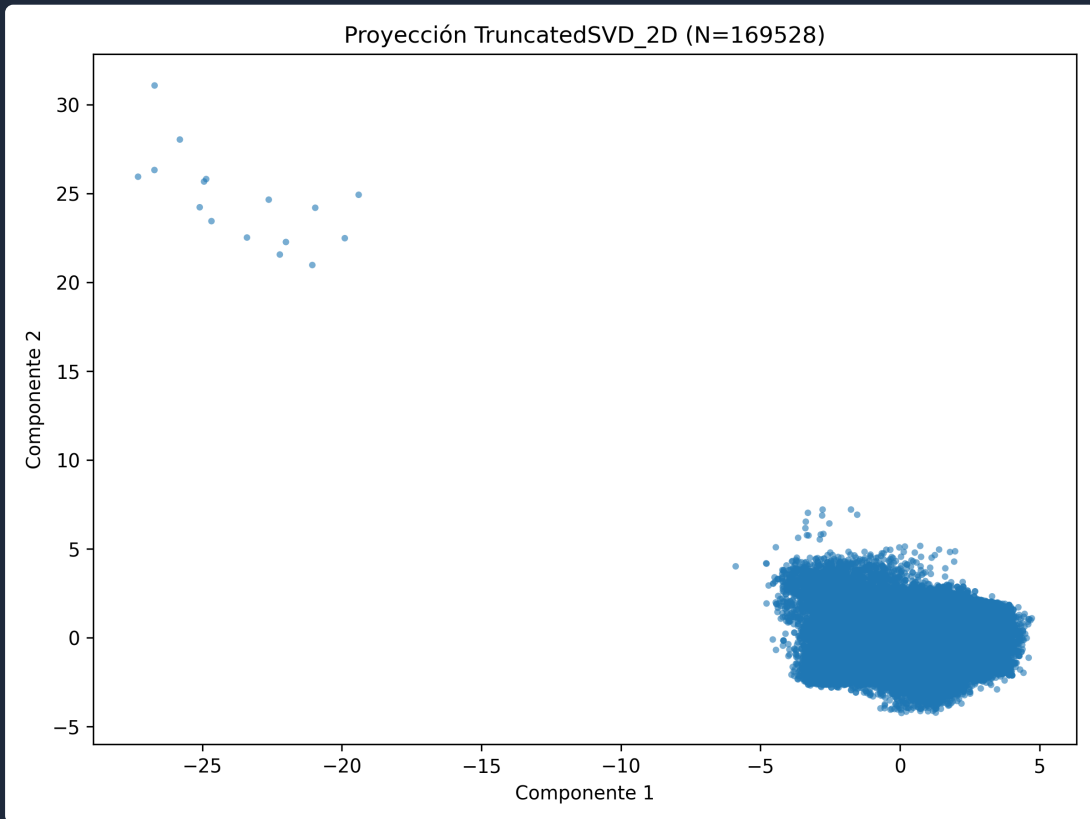


plot_SparsePCA_3D.png

Proyección SparsePCA_3D (N=8000)



plot_TruncatedSVD_2D.png



plot_TruncatedSVD_3D.png

Proyección TruncatedSVD_3D (N=169528)

